

Quadriceps tendon autograft provides comparable stability and functional outcomes with lower donor-site morbidity than bone–patellar tendon–bone: A systematic review and meta-analysis of 15 randomized controlled trials with GRADE evidence

Mohamed Abdo Khalafallah¹  | Omar Khaled Hall¹ | Ziad Elmenawy²  |
Kazhal Shekh Mohamad Ibrahim³ | Omar Mohammad Ghoneim⁴ |
Abdullah Raizah⁵ | Turki Alqahtani⁵ | Mohamed Samir Ahmed Zaki⁵ |
Mohamed A. Imam⁶

¹Faculty of Medicine, Alexandria University, Alexandria, Egypt

²University Hospitals of Dorset, Orthopedic department, Bournemouth, UK

³Medical Laboratory Science, University of Raparin, Ranya, Iraq

⁴Faculty of Medicine, Al Azhar University, Cairo, Egypt

⁵College of Medicine, King Khalid University, Abha, Saudi Arabia

⁶Rowley Bristow Unit, Trauma and Orthopaedics, Ashford and St Peter's Hospitals National Health Service (NHS) Foundation Trust, Lyne, Chertsey, UK

Correspondence

Mohamed Abdo Khalafallah, Faculty of Medicine, Alexandria University, Champollion St, Alexandria 5372066, Egypt.
Email: mohamedabdokhalafallah951@gmail.com

Abstract

Purpose: To compare clinical and functional outcomes after primary anterior cruciate ligament reconstruction (ACLR) using quadriceps tendon (QT) autografts versus bone–patellar tendon–bone (BPTB) and, separately, QT versus hamstring tendon (HT) autografts, to inform evidence-based graft selection.

Methods: We performed a Cochrane-guided systematic review and meta-analysis. Searches of PubMed, Scopus, Cochrane Library, SPORTDiscus, Cumulative Index to Nursing and Allied Health Literature (CINAHL), Web of Science and major trial registries through November 2025 identified randomized controlled trials (RCTs) comparing QT with HT or BPTB in adults undergoing primary ACLR. Random-effects models (REML) were used to pool mean differences/standardized mean differences and risk ratios. Heterogeneity was quantified using I^2 and Q statistics. Sensitivity analyses (leave-one-out) and the GRADE (Grading of Recommendations, Assessment, Development and Evaluation) approach were applied. When pooling was infeasible, a structured narrative synthesis was provided.

Results: Fifteen RCTs (2014–2025) were included. Objective anterior stability (KT-1000) showed no significant difference between QT and HT at 12 or 24 months and no significant difference between QT and BPTB at 12 months. Patient-reported outcomes (International Knee Documentation Committee, Lysholm, Knee Injury and Osteoarthritis Outcome Score [KOOS], Tegner) were largely equivalent at 12–24 months. After resolving inconsistencies due to non-exchangeable constructs/comparators, a small

Abbreviations: ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; AP, anteroposterior; BMI, body mass index; BPTB, bone–patellar tendon–bone; CI, confidence interval; CSA, cross-sectional area; GRADE, Grading of Recommendations, Assessment, Development and Evaluation; HQ, hamstrings-to-quadriceps ratio; HT, hamstring tendon; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; KT-1000, knee laxity testing device; LSI, limb symmetry index; MD, mean difference; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PROMs, patient-reported outcome measures; QT, quadriceps tendon; RCT, randomized controlled trial; RoB 2, Revised Cochrane Risk-of-Bias Tool for Randomized Trials; RR, risk ratio; SD, standard deviation.

12-month KOOS signal favoured QT, while 24-month effects remained null. Graft failure rates were comparable for QT versus HT and QT versus BPTB. Donor-site morbidity was substantially lower with QT versus BPTB; QT versus HT showed no significant difference. Pivot-shift favoured QT, but not significantly; postoperative stiffness did not differ. Overall certainty was mostly moderate (downgrades for imprecision and occasional indirectness); donor-site morbidity versus BPTB reached high certainty.

Conclusion: Across randomized evidence up to 24 months, QT provides stability and patient-reported outcomes comparable to HT and BPTB, with clearly lower donor-site morbidity compared to BPTB and no excess failure compared to HT. QT is an evidence-supported alternative for primary ACLR; longer, adequately powered, construct-specific RCTs are needed for guideline-level certainty.

Level of Evidence: Level I, systematic review of Level I randomized controlled studies.

KEYWORDS

anterior cruciate ligament reconstruction, bone–patellar tendon–bone, hamstring tendon, meta-analysis, quadriceps tendon

INTRODUCTION

Rupture of the anterior cruciate ligament (ACL) commonly necessitates surgical reconstruction to restore knee stability and facilitate return to athletic activity, particularly in young and active patients [20]. A critical determinant of postoperative outcomes is the selection of the autograft used for reconstruction.

The bone–patellar tendon–bone (BPTB) autograft has long been considered a reference standard due to its reliable fixation and low graft failure rates [21]. However, its use has been consistently associated with donor-site morbidity, particularly anterior knee pain and kneeling discomfort, which has contributed to the increasing adoption of hamstring tendon (HT) autografts [21, 32, 46]. Despite their widespread use, HT grafts have demonstrated higher revision rates in several large registry studies, especially among young, high-demand patients [28]. More recently, the quadriceps tendon (QT) autograft has emerged as a viable alternative for primary ACL reconstruction. Advocates highlight its favourable biomechanical properties, predictable graft size and potentially lower donor-site morbidity compared with BPTB and HT grafts [33, 34, 36, 50]. Recent systematic reviews suggest that QT autografts provide comparable stability and graft survival to BPTB, with reduced anterior knee pain and may demonstrate lower graft failure rates than HT autografts while maintaining similar patient-reported outcomes [25, 49].

Despite this growing body of evidence, uncertainty remains regarding optimal graft choice. Many previous systematic reviews are limited by clinical heterogeneity, inclusion of outdated studies, small randomized controlled trials (RCTs) or by pooling BPTB and HT

comparators, thereby obscuring meaningful graft-specific differences. Importantly, contemporary high-quality meta-analyses that separately compare QT versus BPTB and QT versus HT using randomized evidence remain limited. Therefore, the purpose of this systematic review and meta-analysis was to compare clinical, functional and donor-site morbidity outcomes of primary ACL reconstruction using QT autografts versus BPTB autografts and, in a separate analysis, QT autografts versus HT autografts. By synthesizing evidence exclusively from RCTs and applying GRADE (Grading of Recommendations, Assessment, Development and Evaluation) methodology, this study aims to provide a clear, evidence-based framework to guide graft selection in contemporary clinical practice.

METHODS

We conducted this systematic review and meta-analysis in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines and the Cochrane Handbook for Systematic Reviews of Interventions [13, 27]. The study protocol was registered prospectively in PROSPERO (Prospero registration ID: CRD420251182386).

Data sources and search strategy

A comprehensive literature search was performed across PubMed, Scopus, Cochrane Library, SPORT-Discus, CINAHL (Cumulative Index to Nursing and Allied Health Literature), Web of Science and major clinical trial

registries for all studies published up to November 2025. The search strategy was broad and unrestricted by language or publication year to ensure full coverage of relevant evidence. Detailed database-specific search strategies are presented in Appendix S1. All retrieved records were imported into EndNote software for duplicate removal before screening.

Eligibility criteria and study selection

Eligible studies were RCTs enrolling adult patients undergoing anterior cruciate ligament reconstruction (ACLR), in which the QT autograft served as the intervention and the HT or BPTB autografts served as comparators.

We excluded observational studies, case reports, narrative reviews, conference abstracts lacking complete data, biomechanical cadaveric studies and trials involving non-adult participants, multi-ligament reconstructions, revision ACLR or allografts.

Two reviewers independently screened all titles and abstracts using the Rayyan web application [51]. Full-text assessments were conducted for potentially eligible studies, and discrepancies were resolved through discussion or consultation with a third reviewer. We also performed backward and forward citation tracking of included studies to identify additional eligible trials. Reasons for full-text exclusions were documented systematically.

Data extraction

Two reviewers independently extracted data using a standardized and piloted Microsoft Excel sheet designed for this review. Any discrepancies were resolved through discussion and consensus. When any information was unavailable, it was recorded as 'not reported'.

Extracted information included:

- Study characteristics: author, publication year, country, inclusion and exclusion criteria, details of the intervention and comparator, total sample size, duration of follow-up and outcomes assessed.
- Baseline demographics: age, sex, body mass index (BMI), height, weight and baseline pain score (Visual Analogue Scale [VAS] or equivalent). Continuous variables were extracted as mean $\pm$ standard deviation (SD), and categorical variables as number and percentage (N [%]) where available.

Outcomes and definitions

The primary and secondary outcomes were selected based on their clinical relevance and frequent reporting

across included trials evaluating QT, HT and BPTB autografts for ACLR. Objective knee stability was primarily assessed using the Knee laxity testing device (KT-1000) arthrometer, which quantifies anterior tibial translation (side-to-side difference in millimetres) between the operated and contralateral limbs. Functional outcomes were evaluated using validated, patient-reported instruments, including the International Knee Documentation Committee (IKDC) subjective knee score, Lysholm Knee Scoring Scale, Knee Injury and Osteoarthritis Outcome Score (KOOS) and Tegner Activity Scale, each of which was measured at 12- and 24-month follow-up intervals depending on study availability. Pain intensity was assessed using the VAS, while graft failure was defined as either revision ACLR or a confirmed graft rupture verified clinically or radiologically.

Donor-site morbidity encompassed complications directly attributable to graft harvest, including anterior knee pain, kneeling pain, hypaesthesia or quadriceps weakness, reported as dichotomous (event-based) outcomes where available. Knee stability was further evaluated using the pivot-shift test, reported as the proportion of patients demonstrating a positive pivot response postoperatively. Additionally, postoperative stiffness was defined as a limitation of the knee's range of motion that necessitated intervention.

Risk of bias assessment

Two reviewers independently evaluated the methodological quality of all included studies. The included RCTs were assessed using the Revised Cochrane Risk of Bias Tool (RoB 2) [38], which examines potential bias across five key domains: (D1) the randomization process, (D2) deviations from intended interventions, (D3) completeness of outcome data, (D4) accuracy of outcome measurements and (D5) selection of reported results. Agreement between reviewers was quantified using the Kappa statistic [38], with values interpreted as follows: 0.81–1.00 indicating almost perfect agreement, 0.61–0.80 indicating substantial agreement, 0.41–0.60 indicating moderate agreement, 0.21–0.40 indicating fair agreement and 0.00–0.20 indicating slight agreement.

Statistical analysis

All quantitative analyses were performed using Stata Version 17.0 (StataCorp). Effect estimates were expressed as mean difference (MD) for continuous variables and risk ratio (RR) for dichotomous variables, each with corresponding 95% confidence intervals (CIs). When continuous outcomes were measured using different scales across studies or the pooled

effect size was heterogeneous, results were summarized using the standardized mean difference (SMD). A random-effects model with the restricted maximum likelihood (REML) estimator was applied to account for between-study variability. A two-tailed p value of <0.05 was considered statistically significant for pooled effect estimates.

For studies containing zero-event cells, a continuity correction of 0.5 was applied to enable computation of pooled RRs. Statistical heterogeneity was assessed using the I^2 statistic and Cochran's Q test, with $p < 0.10$ indicating significant heterogeneity. Separate meta-analyses were pre-specified and conducted for QT versus HT and QT versus BPTB comparisons. Pooling of HT and BPTB autografts into a combined comparator group (QT vs. HT/BPTB) was performed only when subgroup-specific data were insufficient to generate statistically stable estimates (e.g., when a single study represented one comparator subgroup for a given outcome). These pooled analyses were considered supplementary and interpreted cautiously to preserve graft-specific interpretability. The robustness of pooled estimates was examined through leave-one-out sensitivity analyses across all outcomes, ensuring that no single study unduly influenced the overall effect.

In outcomes where a single study represented the BPTB subgroup, sensitivity testing confirmed that exclusion of this study did not materially alter the pooled estimates, supporting the stability of the overall findings. Owing to the limited number of studies (<10) per outcome, publication bias could not be formally assessed using funnel plot asymmetry or Egger's regression [8]. The certainty of evidence for each pooled outcome was evaluated using the GRADE approach, considering the risk of bias, inconsistency, indirectness, imprecision and publication bias domains [10, 11, 26].

Narrative synthesis

When data could not be pooled due to inconsistent reporting formats, heterogeneity in measurement tools or missing statistical data, we conducted a narrative synthesis in accordance with Cochrane and PRISMA 2020 guidelines. Findings were systematically summarized to describe trends in biomechanical performance, functional recovery, muscle strength symmetry and postoperative complications across QT, HT and BPTB autografts. Results were grouped thematically by outcome domain (e.g., instrumented laxity, strength ratios, patient-reported outcomes, surgical complications) and, when possible, by follow-up duration. Where direct comparisons were available, differences were interpreted qualitatively based on the direction of the effect and statistical significance. Studies reporting p values without variance measures were described

narratively but excluded from meta-analysis to maintain methodological rigour.

RESULTS

Study selection

The initial search retrieved 5463 studies. After removing 3200 duplicates, 2263 studies were processed for title and abstract screening. After excluding 2209 studies, the remaining 54 studies were processed for full-text screening. After excluding ten reviews and 29 observational studies, 15 RCTs were included in the final synthesis. The screening process is presented in Figure 1.

Study characteristics and demographics

The included studies comprised 15 comparative investigations conducted between 2014 and 2025 across multiple regions, including India, Mexico, Australia, Turkey, Germany, Denmark, Spain and Romania. All included RCTs were published in English. Eligible participants were generally young to middle-aged adults (16–50 years). Sample sizes ranged from 28 to 116 participants per trial, with follow-up durations spanning from early postoperative assessments to up to 10 years. A summary of the common characteristics of the included studies is presented in Table 1, and baseline characteristics are presented in Table 2.

Quality assessment

Using the RoB 2 tool (Figure 2), 14 studies were judged as having some concerns overall, while one study was rated as high risk. In Domain 1, one study demonstrated some concerns, whereas 14 were rated as low risk. Regarding Domain 2, nine studies were judged to be of low risk, five had some concerns, and one was considered high risk, primarily due to the inherent limitations of blinding participants and surgeons in surgical procedures. For Domain 3, 12 studies were low risk, and three showed some concerns. In Domain 4, 14 studies were rated as low risk, and one was rated as high risk. Finally, in Domain 5, one study was judged to be of low risk. At the same time, 14 studies expressed some concerns, reflecting incomplete reporting or insufficient clarity in the predefined outcome measures. Despite these limitations, the overall evidence base was considered to have a moderate risk of bias, consistent with typical challenges in surgical RCTs where blinding is difficult to maintain. The interrater reliability between reviewers was high, with a Kappa coefficient of 0.90, indicating almost perfect agreement.

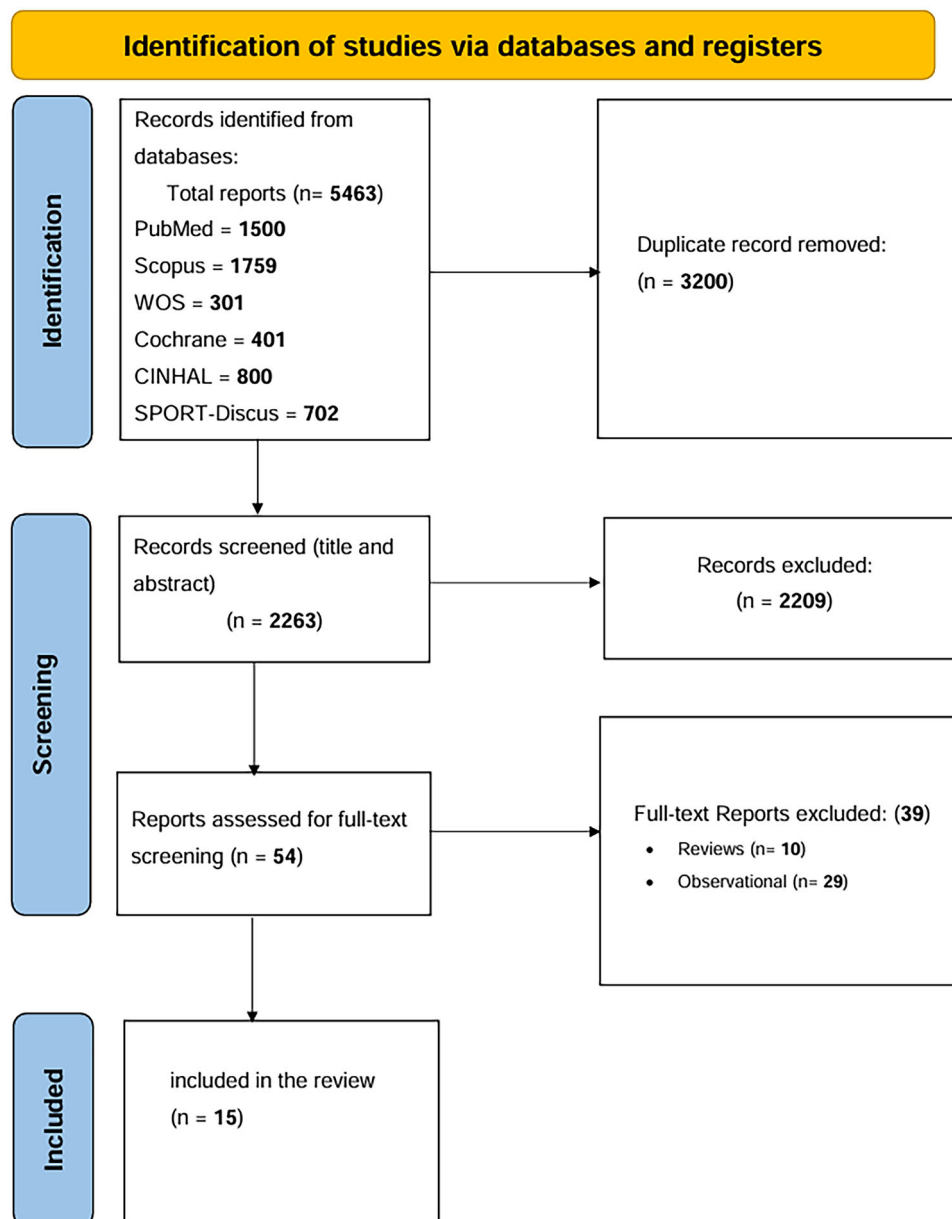


FIGURE 1 PRISMA 2020 flow diagram of study selection. Flow diagram illustrating the systematic search, screening, eligibility assessment and final inclusion of randomized controlled trials comparing quadriceps tendon (QT) with hamstring tendon (HT) and bone–patellar tendon–bone (BPTB) autografts for primary ACL reconstruction. ACL, anterior cruciate ligament; CINHAL, Cumulative Index to Nursing and Allied Health Literature; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; WOS, Web of Science.

Results of the meta-analysis

KT-1000 side-to-side difference

Three studies [7, 17, 48] including 261 participants (QT = 128; HT = 133) evaluated side-to-side difference at 12 months. The pooled analysis demonstrated no statistically significant difference between groups (SMD = −0.20; 95% CI −0.55 to 0.14; $p = 0.24$; $I^2 = 49\%$; $p_{\text{het}} = 0.14$) (Figure 3a). Sensitivity analysis using a leave-one-out approach confirmed the robustness of the pooled estimate.

Two additional studies [7, 14] (QT = 69; HT = 75) at 24 months reported similar findings, with no significant difference between graft types (MD = −0.15 mm; 95% CI −0.51 to 0.21; $p = 0.42$; $I^2 = 0\%$; $p_{\text{het}} = 0.48$) (Figure 3b).

A further two studies [3, 19] (QT = 54; BPTB = 52) showed no significant difference at 12 months (MD = 0.16 mm; 95% CI −0.54 to 0.85; $p = 0.66$; $I^2 = 52\%$; $p_{\text{het}} = 0.15$) (Figure 3c). Collectively, these findings indicate comparable anterior knee stability between QT, HT and BPTB autografts.

TABLE 1 Characteristics of included studies.

Study ID	Country	Inclusion	Exclusion criteria	Intervention details	Comparator details	Sample size	Follow-up	Outcomes measurement	Findings
Yadav et al., 2025	India	Patients aged 18–45 with MRI-proven complete ACL tear and no other ligament injury; partial meniscectomy/meniscal repair allowed.	Revision ACL cases, prior fractures, refusal to participate.	Partial-thickness QT autograft harvested minimally invasively (2 cm strip, folded, 6.6–7 cm × 7.5–10 mm), all-inside single-bundle arthroscopic reconstruction.	HT autograft (semitendinosus + gracilis), quadrupled, 6.6–7 cm × 7.5–10 mm, same technique.	69 (QT = 33; HT = 36)	6 and 12 months	Lachman, anterior drawer, pivot shift, KT-1000; VAS; Lysholm, KOOS; graft failure, donor-site morbidity, PROM differences small; QT favoured for rotational stability; long-term validation needed.	QT showed better translational stability + lower donor-site morbidity; PROM differences small; QT favoured for rotational stability; long-term validation needed.
García-Linage et al., 2025	Mexico	Age 18–40, Tegner ≥ 4, Grade III ACL rupture, managed by study surgeon.	Previous knee surgery, illicit drug use, inability for 360-day follow-up, degenerative/genetic disease, meniscal/chondral lesions needing NWB.	QT ≈ 10 mm × 6.5 cm, all-inside, button fixation, no brace, physio Day 10.	HT (semitendinosus), same surgical and rehab protocol.	32 (6 excluded for conversion/loss)	30, 180, 360 days	VAS, Lysholm, mCKRS, ACL-RSI, Tegner, return-to-sport time, rehab days, stability tests.	QT had higher early pain (VAS 30/180 days) but equal by 360 days; function and stability similar; QT better early psychological readiness.
Calvert et al., 2025	Australia	ACLR candidates (16–50 years) without meniscal pathology, MRI confirmed.	BMI ≥ 40, revision/multiligament cases, cartilage grade ≥ 3, psychiatric or substance abuse.	All-soft-tissue QT autograft (70 mm, 9–11 mm harvester) per Arthrex Graftlink protocol, adjustable loop fixation.	HT (semitendinosus ± gracilis) quadrupled ≥ 8 mm, same fixation/protocol.	112 (97 analysed)	6, 12, 24 months	Kneeling tolerance (KT90/110), IKDC, Lysholm, ACL-RSI, Global Rating of Change.	QT autograft yielded superior kneeling tolerance up to 2 years post-op versus HT.
Tang et al., 2024	Turkey/Istanbul	Isolated primary ACL injury (18–50 years), no prior surgery.	Multi-ligament injury, revision, Outerbridge > 2, age <18 or >50.	Bone-free QT autograft via 4 cm longitudinal incision.	Hamstring autograft via 4 cm oblique incision medial proximal tibia.	40 (from 64 initial cases)	3, 6, 9, 12, 24 months	Stability tests, strength testing, IKDC, KOOS, Lysholm scores.	Both groups improved; QT grafts larger; HT had more saphenous nerve palsy.

TABLE 1 (Continued)

Study ID	Country	Inclusion	Exclusion criteria	Intervention details	Comparator details	Sample size	Follow-up	Outcomes measurement	Findings
Ebert et al., 2024	Australia	Age 16–50, clinically/MRI confirmed ACL rupture, eligible for primary ACLR.	Refusal, BMI ≥ 40 , revision/multiligament, cartilage pathology, psychiatric or substance abuse.	QT all-soft tissue (~70 mm), Arthrex Grafliink system with adjustable loops, same surgeon.	HT (semitendinosus $\pm$ gracilis), same system & protocol.	116 $\rightarrow$ 112 operated $\rightarrow$ 97 completed (HT 49, QT 48)	6 weeks, 3, 6, 12, 24 months	KT-1000, IKDC, Lysholm, Cincinnati, Tegner, ACL-RSI, Donor Site Score, Pain VAS, isokinetic strength, hop tests.	PROMs improved in both; HT higher ACL-RSI early; no laxity difference; HT greater quad strength 6–12 months; QT better hamstring strength; comparable return to sport and reinjury.
Horstmann et al., 2022	Germany	ACL insufficiency (age 18–50) confirmed clinically + MRI.	Revision, PCL or PLC injury, $> \frac{1}{2}$ meniscus loss, full-thickness cartilage lesion, fracture.	QT with patellar bone block, press-fit fixation (no screw), standard rehab.	HT (four-strand ST/G), same press-fit and rehab.	51 (26 HT, 21 QT analysed)	3, 6, 12, 24 months	KT-1000, IKDC, Lysholm, pivot shift, isokinetic strength, return to sport/work, complications.	Both improved; no sig. group differences; return to sport $\approx$ 95 days (HT) versus 82 days (QT); QT 3 retears versus 1 HT.
Vilchez-Cavazos et al., 2020	Mexico	Adults with clinical + MRI ACL rupture ($\pm$ meniscus), no prior surgery.	Multiligament, OA, degenerative/rheumatoid disease, contralateral ACL injury, lost to follow-up.	QT autograft per Rabuck et al. technique.	HT harvest per Petersen and Zantop method.	28 (14 per group; 3 excluded)	Baseline; 2 weeks; 1, 3, 6, 12 months	VAS, Lysholm, IKDC, SF-12, ROM.	QT and HT had similar pain and function outcomes at all times.
Sinding et al., 2020	Denmark	≥ 18 years, isolated ACL-R, treated 2014–2016 at Aarhus Hospital.	Diabetes, arthritis, malignancy, other instability, ankylosing spondylitis, BMI > 30 , glucocorticoids, rehab inability.	QT (partial-thickness 5 mm with 20 mm bone block).	ST/G four-strand hamstring graft.	100 randomized (42 QT, 43 STG final) + 50 healthy controls.	12 months	Isokinetic KE/KF strength, LSI, HQ ratio, Single Leg Hop, IKDC.	STG $\rightarrow$ both KE and KF weakness; QT $\rightarrow$ KE only; function (IKDC, SHD) similar, both weaker than controls.

(Continues)

TABLE 1 (Continued)

Study ID	Country	Inclusion	Exclusion criteria	Intervention details	Comparator details	Sample size	Follow-up	Outcomes measurement	Findings
Lind et al., 2020	Denmark	Adults with isolated ACL injuries (MRI + instability symptoms).	Malignancy, other instability, RA, Bechterew, BMI > 30, glucocorticoids, rehab inability.	QT (5 mm partial thickness + 20 mm bone block).	ST/G four-strand hamstring graft.	99 (QT 50; STG 49)	1 and 2 years	IKDC (primary); KOOS, Kujala, Tegner, KT-1000, pivot shift, one-leg hop, donor site morbidity.	No difference in IKDC/ KOOS; QT less donor-site pain ($p = 0.046$); QT lower hop symmetry (91% vs. 97%); stability similar.
Barié et al., 2020	Germany	Primary ACL rupture; Tegner ≥ 6 .	ACL re-rupture, PCL/lateral instability, subtotal meniscectomy, OA Grade 3–4, Outerbridge > 2.	QT autograft press-fit (no hardware) bone-to-bone healing.	BPTB autograft press-fit (no hardware).	60 (30 per group); 43 at 10 years.	1 and 10 years	Lysholm, IKDC, Tegner, VAS satisfaction, KT-1000, X-ray tunnels, complications.	No significant functional differences; BPTB had higher kneeling/squatting pain at 1 and 10 years.
Tirupathi et al., 2019	India	Age 20–50, isolated ACL injury (single-bundle QT or HT), MRI confirmed.	Prior surgery, knee deformity, major comorbidity.	QT autograft (9×100 mm), IFS (Megafix Storz) fixation.	HT (semitendinosus + gracilis), IFS fixation.	96 (10 lost)	3 weeks–24 months	IKDC, stability (Lachman/ drawer/pivot), strength (quad/ham), balance, complications.	Both stable and comparable at 2 years; QT temporary extensor weakness, HT temporary flexor weakness.
Martin-Alguacil et al., 2019	Spain	Confirmed ACL rupture, <6 months since injury, athletes.	Prior injury/ surgery, concomitant ligament/ meniscus injury, severe cartilage lesion.	QT free bone plug ($80 \times 10 \times 7$ mm), bioabsorbable tibial fixation.	HT (semitendinosus + gracilis) 3 cm oblique incision, suspensory femoral fixation.	56 (31 final)	3, 6, 12 months	Pain (PPT via algometer), muscle architecture differences; hop test, isokinetic strength, laxity.	No significant pain or architecture differences; both recovered pain sensitivity by 1 year.
Martin-Alguacil et al., 2018	Spain	ACL injury <6 months, competitive soccer players.	Prior surgery, other ligament injury, >50% meniscus resected, cartilage > Grade II.	QT (no bone block) 4 cm vertical incision $70\text{--}80 \times 10 \times 7$ mm, bioabsorbable fixation.	HT (ST $\pm$ G) 4 cm oblique incision, 5-strand for <7 mm, suspensory + bio screw fixation.	56 (51 final)	3, 6, 12, 24 months	H/Q ratio, peak torque, Lysholm, Cincinnati, KT-2000, re-rupture.	QT had higher H/Q ratio (3–12 months); no functional difference at 24 mo; trend toward fewer re-ruptures versus HT.

TABLE 1 (Continued)

Study ID	Country	Inclusion	Exclusion criteria	Intervention details	Comparator details	Sample size	Follow-up	Outcomes measurement	Findings
Buescu et al., 2017	Romania	Age 16–50, scheduled for arthroscopic ACLR.	Meniscus repair needed, multi-ligament, prior surgery, chronic pain, hepatic disease, substance abuse, recent analgesics.	Free QT autograft (4 cm midline), extracortical femoral button, bioabsorbable tibial screw.	Hamstring autograft (3 cm anteromedial incision), same fixation.	48	0–48 h post-operative	VAS pain, rescue analgesia, timing, complications.	HT group required 38% more analgesic; QT group 50% needed no rescue (p = 0.002); no complications.
Lund et al., 2014	Denmark	Isolated adult ACL injuries (>18 years), surgeon initially chose BPTB graft.	Previous reconstruction, multi-ligament, bilateral, cartilage >2 cm ² /50% depth, major meniscus repair.	QT with bone plug (20 × 9–10 mm), 7 × 25 mm femoral screw, Intrafix tibial fixation.	BPTB (central 10 mm), 7 × 25 mm femoral + 9 × 25 mm tibial screws, bone defect filled.	51 analysed (of 66 enroled)	12 and 24 months	KT-1000, pivot shift, kneeling pain, harvest pain, tibial sensitivity loss, KOOS, IKDC.	QT had less kneeling/harvest pain & sensitivity loss with better rotational stability; similar laxity and scores to BPTB.

Note: Summary of study characteristics including country, sample size, intervention and comparator graft type, follow-up duration and reported outcomes.

Abbreviations: ACL, anterior cruciate ligament; ACLR, ACL reconstruction; ACL-RSI, ACL return-to-sport after injury; BMI, body mass index; BPTB, bone-patellar tendon-bone autograft; GRC, Global Rating of Change; HT, hamstring tendon autograft (semitendinosus ± gracilis); HQ, hamstring-to-quadriceps ratio; IKDC, International Knee Documentation Committee; KE/KF, Knee Extensor/Flexor; KOOS, Knee Injury and Osteoarthritis Outcome Score; KT-1000, Arthrometer for anterior tibial translation; LSI, Limb Symmetry Index; mCKRS, Modified Cincinnati Knee Rating System; MRI, magnetic resonance imaging; ns, not significant; NWB, non-weight bearing; PCL, posterior cruciate ligament; PPT, pain pressure threshold; PROMs, patient-reported outcome measures; QT, quadriceps tendon autograft; QTB, quadriceps tendon with patellar bone block; RA, rheumatoid arthritis; ROM, range of motion; RTS, Return to Sport; SF-12, 12-item Short-Form Health Survey; SHD, single hop for distance; ST/G, semitendinosus/gracilis graft; Tegner, Tegner Activity Scale; VAS, Visual Analogue Scale.

TABLE 2 Baseline demographic characteristics.

Study ID	Group	N	Age, years Mean (SD), range	BMI Mean (SD)	Weight (kg) Mean (SD)	Height (cm) Mean (SD)	Pain score (VAS or others) Mean (SD)	Male N (%)
Yadav et al., 2025	QT	33	28.73 (8.80)	—	—	—	1.58 (0.79)	30 (90.9)
	HT	36	30.06 (8.87)	—	—	—	1.92 (1.27)	32 (88.9)
García-Linage et al., 2025	QT	13	26.23 (6.2)	23.99 (1.8)	—	—	—	6 (46.15)
	HT	13	28.46 (7.19)	24.00 (2.11)	—	—	—	7 (53.85)
Calvert et al., 2025	QT	57	28.1 (8.2)	26.6 (3.6)	—	—	—	28 (49.1)
	HT	55	29.4 (7.7)	26.3 (3.6)	—	—	—	28 (50.9)
Tang et al., 2024	QT	17	28.06 (6.24)		77.35 (13.54)	175.65 (5.95)		17 (100)
	HT	16	28.31 (8.55)		77.56 (16.93)	173.44 (7.58)		13 (81.25)
Ebert et al., 2024	QT	57	28.1 (8.2)	26.6 (3.6)	—	—	—	28 (49.1)
	HT	55	29.4 (7.7)	26.3 (3.6)	—	—	—	28 (50.9)
Horstmann et al., 2022	QT	24	24.1 (3.6)	24.9(3.8)	79.0 (13.6)	177.9 (7.3)	—	21 (87.5)
	HT	27	32.7 (11.4)	25.2 (4.0)	73.6 (11.3)	171.2 (8.2)	—	12 (44.4)
Vilchez-Cavazos et al., 2020	QT	14	23 (20–30)	—	—	—	4 (3-7)	13 (80)
	HT	14	23 (19.5-30.5)	—	—	—	3 (1.5-7)	11(84.6)
Sinding et al., 2020	QT	42	28.7 (6.4)	25.1 (3.2)	77.0 (11.5)	175 (8.9)	—	25 (60)
	HT	43	28.3 (6.2)	24.3 (3.2)	75.6 (13.0)	176 (8.6)	—	23 (53)
Lind et al., 2020	QT	50	27.2 (6.4)	24.3 (3.4)	—	—	—	29 (58)
	HT	49	27.1 (6.1)	23.8 (2.6)	—	—	—	25 (51)
Barié et al., 2020	QT	30	30.5 (7.8)	—	—	—	—	17 (57)
	BPTB	30	30.6 (7.5)	—	—	—	—	17 (57)
Tirupathi et al., 2019	QT	48	—	—	—	—	—	—
	HT	48	—	—	—	—	—	—
Martin-Alguacil et al., 2019	QT	26	18.7 (3.6)	23.0 (2.2)	—	—	—	23 (88.5)
	HT	25	19.2 (3.6)	23.5 (3.5)	—	—	—	19 (76.0)
Martin-Alguacil et al., 2018	QT	26	18.7 (3.6)	23.0 (2.2)	—	—	—	23 (88.5)
	HT	25	19.2 (3.6)	23.5 (3.5)	—	—	—	16 (64.0)
Buescu et al., 2017	QT	24	29.21 (8.52)	26.27 (4.85)	—	—	—	38 (79.2)
	HT	24	27.50 (5.62)	25.68 (3.32)	—	—	—	
Lund et al., 2014	QT	26	30 (9)	—	—	—	—	21 (81)
	BPTB	25	31 (8)	—	—	—	—	21 (84)

Note: Baseline patient characteristics across included studies, presented as mean ± SD for continuous variables and number (percentage) for categorical variables. Abbreviations: BMI, body mass index; BPTB, bone–patellar tendon–bone autograft; HT, hamstring tendon; QT, quadriceps tendon; SD, standard deviation; VAS, Visual Analogue Scale.

IKDC scores

Three studies [4, 17, 19] (QT = 126; HT/BPTB = 122) at 12 months showed no significant difference in IKDC scores (MD = −1.64; 95% CI −5.15 to 1.87; $p = 0.36$; $I^2 = 0\%$; $p_{\text{het}} = 0.69$) (Figure 4a).

Four studies [4, 14, 17, 41] (QT = 130; HT = 139) at 24 months also revealed no significant difference (MD = 0.08; 95% CI −4.18 to 4.34; $p = 0.97$; $I^2 = 63\%$; $p_{\text{het}} = 0.06$) (Figure 4b). After exclusion of Horstmann et al., heterogeneity decreased substantially ($I^2 = 38\%$; $p_{\text{het}} = 0.16$) without changing the overall

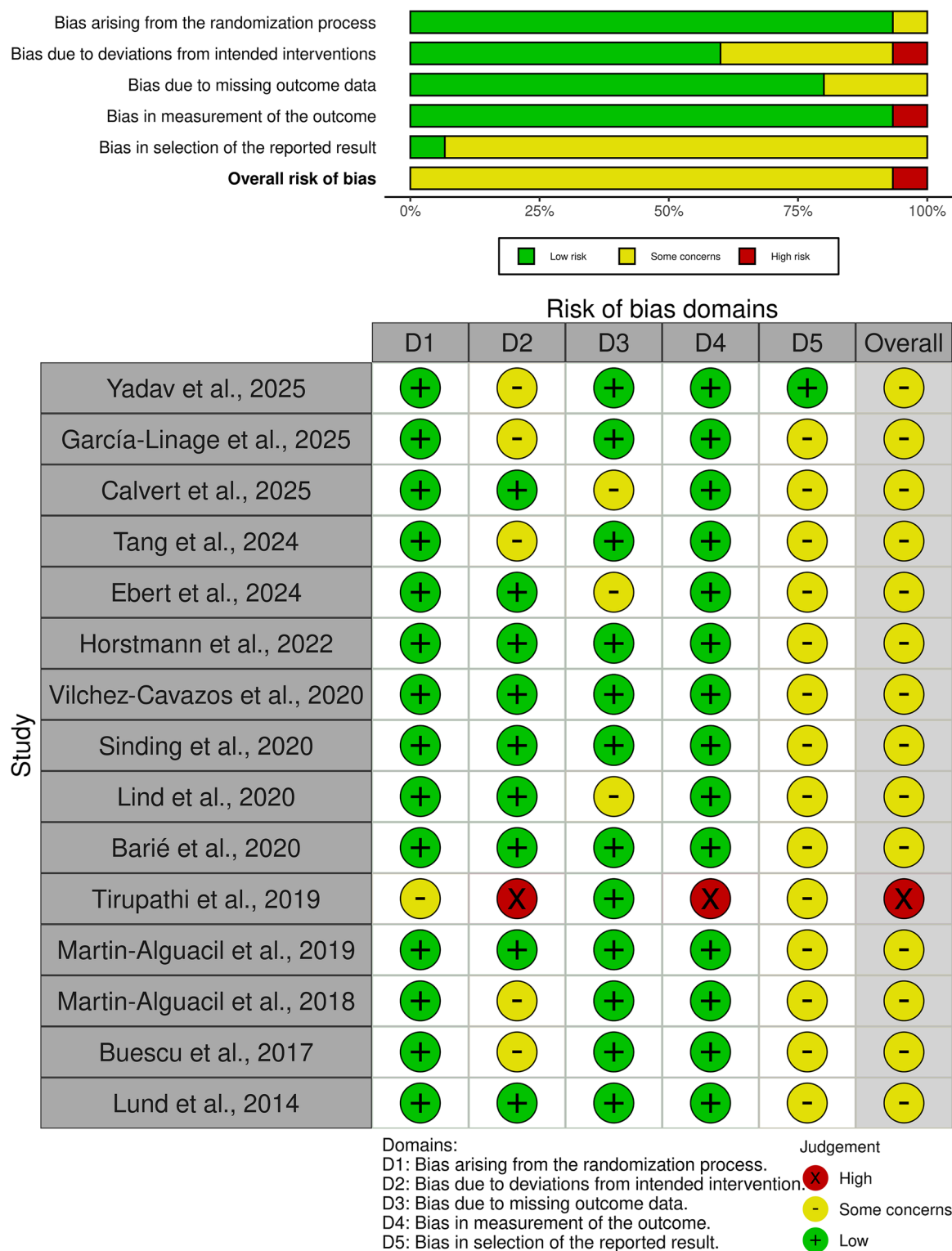


FIGURE 2 Risk of Bias Assessment (RoB 2 Tool). Summary of methodological quality assessment of included randomized controlled trials using the Revised Cochrane Risk of Bias Tool (RoB 2). Domains assessed included: (D1) randomization process, (D2) deviations from intended interventions, (D3) missing outcome data, (D4) measurement of outcomes and (D5) selection of reported results.

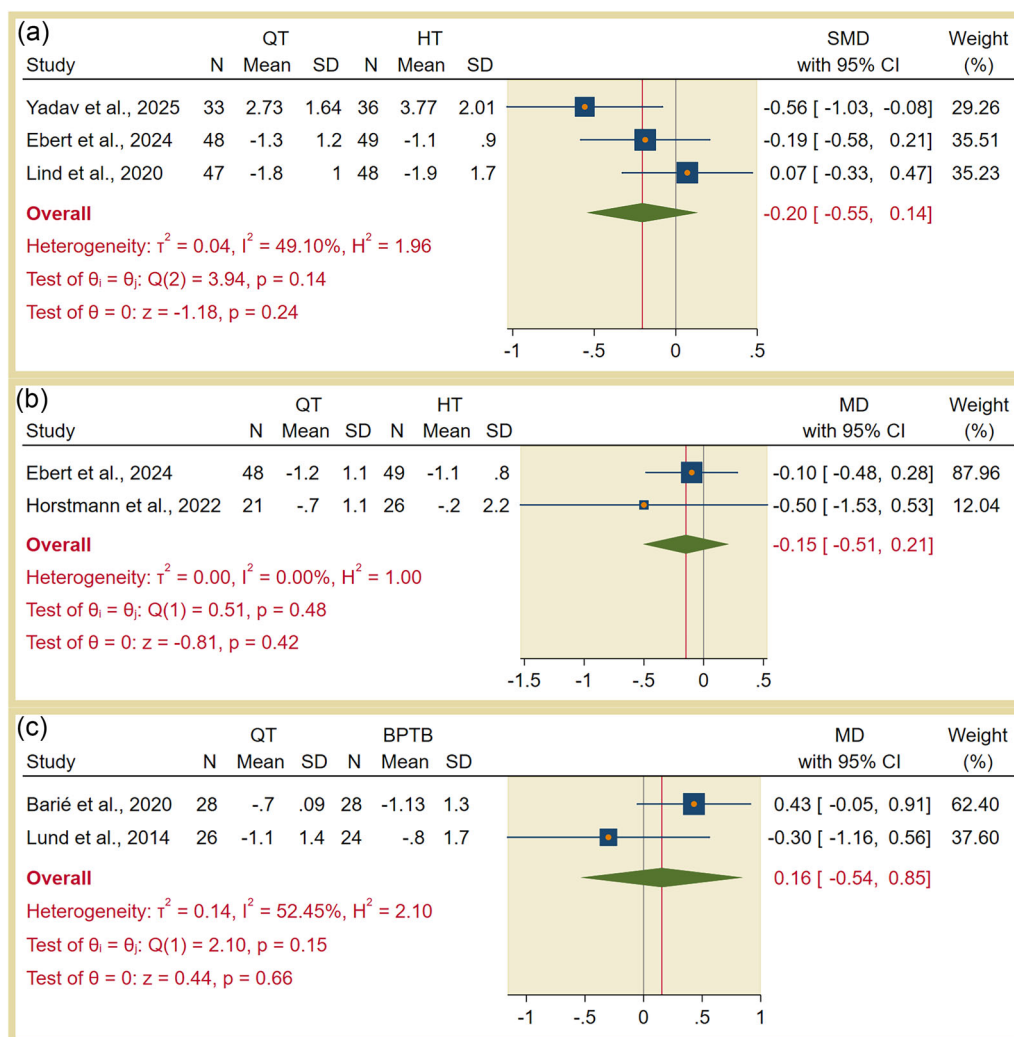


FIGURE 3 KT-1000 side-to-side difference. Forest plots comparing anterior tibial translation (side-to-side difference, mm) between QT and comparator grafts. (a) QT versus HT at 12 months; (b) QT versus HT at 24 months; (c) QT versus BPTB at 12 months. Effect measure: MD or SMD with 95% CIs. Negative values favour QT. BPTB, bone–patellar tendon–bone; CI, confidence interval; HT, hamstring tendon; KT-1000, Knee laxity testing device; MD, mean difference; QT, quadriceps tendon; SD, standard deviation; SMD, standardized mean difference.

result (MD = -1.57; 95% CI -5.01 to 1.87; $p = 0.37$) (Figure 4c).

Lysholm scores

Five studies [3, 4, 9, 44, 48] (QT = 141; HT/BPTB = 140) at 12 months reported no significant difference in functional outcomes (MD = 0.84; 95% CI -1.84 to 3.52; $p = 0.54$; $I^2 = 61\%$; $p_{het} = 0.02$) (Figure 5a). After excluding Yadav et al., heterogeneity resolved ($I^2 = 0\%$; $p_{het} = 0.99$), and the overall estimate remained stable (MD = -0.62; 95% CI -2.73 to 1.48; $p = 0.56$) (Figure 5b).

At 24 months, three studies [4, 14, 41] (QT = 86; HT = 91) demonstrated no significant difference (MD = -0.14; 95% CI -2.46 to 2.19; $p = 0.91$; $I^2 = 0\%$; $p_{het} = 0.25$) (Figure 5c).

KOOS score

Four studies [7, 17, 19, 48] (QT = 159; HT/BPTB = 157) at 12 months reported no significant difference between grafts (MD = 0.12; 95% CI -4.16 to 4.41; $p = 0.96$; $I^2 = 80\%$; $p_{het} < 0.01$) (Figure 6a). After excluding Yadav et al., heterogeneity resolved ($I^2 = 0\%$; $p_{het} = 0.96$), and QT showed slightly better outcomes (MD = -2.48; 95% CI -4.71 to -0.26; $p = 0.03$) (Figure 6b).

At 24 months, four studies [7, 17, 19, 41] (QT = 130; HT/BPTB = 130) demonstrated no significant difference (MD = 0.36; 95% CI -3.60 to 4.31; $p = 0.86$; $I^2 = 75\%$; $p_{het} = 0.06$) (Figure 6c). Excluding Lund et al. reduced heterogeneity ($I^2 = 1\%$; $p_{het} = 0.15$) with marginally improved favorability for QT (MD = -1.17; 95% CI -2.41 to 0.07; $p = 0.06$) (Figure 6d).

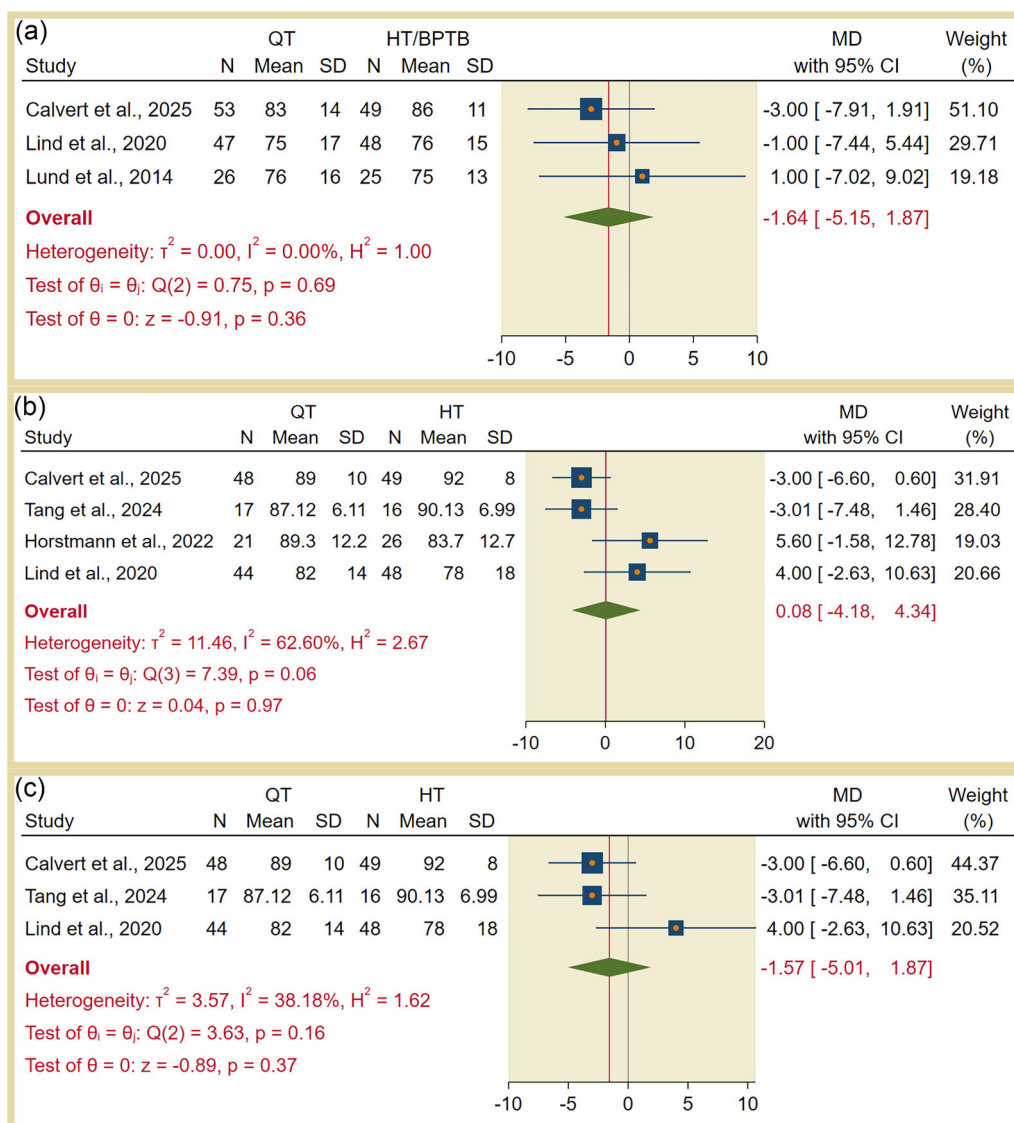


FIGURE 4 IKDC subjective scores. Forest plots comparing IKDC subjective knee scores between QT and comparator grafts. Higher scores indicate better knee function. Effect measure: MD with 95% CI. BPTB, bone–patellar tendon–bone; CI, confidence interval; HT, hamstring tendon; IKDC, International Knee Documentation Committee; MD, mean difference; QT, quadriceps tendon; SD, standard deviation. (a) IKDC scores at 12 months (QT vs HT/BPTB). (b) IKDC scores at 24 months (QT vs HT). (c) Sensitivity analysis of 24-month IKDC scores after exclusion of Horstmann et al. [14].

Tegner score

Four studies [3, 7, 9, 17] (QT = 141; HT/BPTB = 138) at 12 months showed no significant difference in activity levels (SMD = -0.22; 95% CI -0.58 to 0.13; $p = 0.22$; $I^2 = 52\%$; $p_{\text{het}} = 0.10$) (Figure 7a). At 24 months, two studies [7, 17] (QT = 92; HT = 96) demonstrated similar results (MD = 0.01; 95% CI -0.41 to 0.43; $p = 0.97$; $I^2 = 0\%$; $p_{\text{het}} = 0.64$) (Figure 7b).

Visual Analogue Scale (VAS) for pain

Two studies [9, 48] (QT = 46; HT = 49) found no significant difference at 12 months (SMD = -0.38; 95%

CI -2.40 to 1.64; $p = 0.71$; $I^2 = 95\%$; $p_{\text{het}} < 0.01$) (Figure 7c).

Graft failure

Five studies [7, 14, 17, 23, 48] (QT = 192; HT = 195) reported comparable graft failure rates (RR = 1.03; 95% CI 0.32–3.30; $p = 0.96$; $I^2 = 0\%$; $p_{\text{het}} = 0.62$) (Figure 8a). Sensitivity analysis confirmed result stability.

Two studies [3, 19] (QT = 56; BPTB = 55) also showed no significant difference (RR = 0.34; 95% CI 0.04–3.16; $p = 0.34$; $I^2 = 0\%$; $p_{\text{het}} = 0.99$) (Figure 8b).

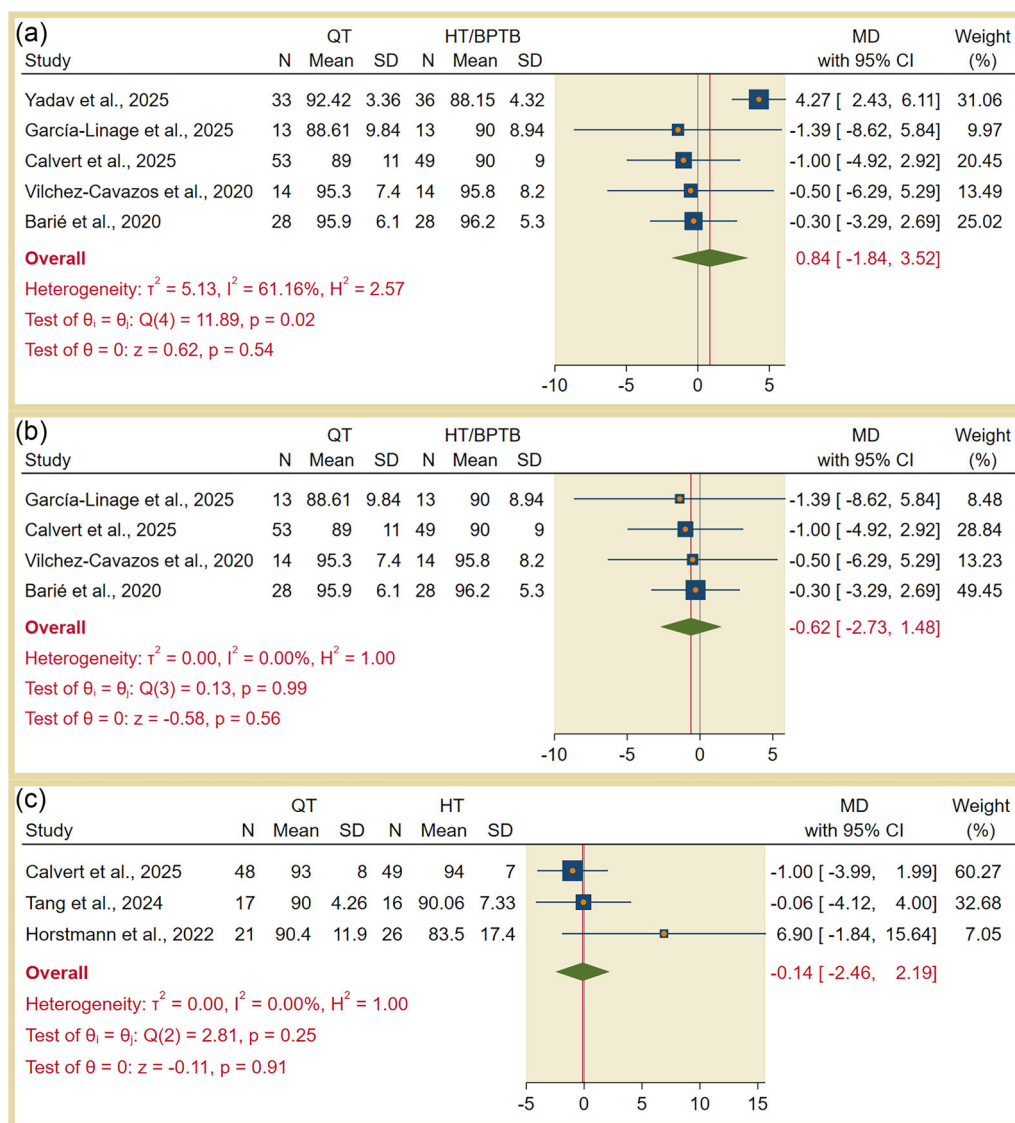


FIGURE 5 Lysholm scores. Forest plots demonstrating pooled MDs in Lysholm functional scores between QT and comparator grafts at 12 and 24 months. BPTB, bone–patellar tendon–bone; CI, confidence interval; HT, hamstring tendon; MD, mean difference; QT, quadriceps tendon; SD, standard deviation. (a) Lysholm scores at 12 months. (b) Sensitivity analysis of 12-month Lysholm scores after exclusion of Yadav et al. [48]. (c) Lysholm scores at 24 months.

Donor site morbidity

Two studies [17, 48] (QT = 83; HT = 85) reported no significant difference (RR = 0.30; 95% CI 0.03–2.62; $p = 0.28$; $I^2 = 62\%$; $p_{het} = 0.11$) (Figure 8c).

Two studies [3, 19] (QT = 56; BPTB = 54) revealed significantly lower donor-site morbidity in the QT group (RR = 0.07; 95% CI 0.01–0.51; $p = 0.01$; $I^2 = 0\%$; $p_{het} = 0.98$) (Figure 8d).

Two studies were excluded from the pooled analysis [7, 44] because they reported donor site morbidity as continuous data rather than specifying event frequencies. Both studies compared QT and HT autografts and found no significant

differences in donor site morbidity within 12 months postoperatively.

Pivot-shift test

Five studies [9, 14, 17, 19, 48] (QT = 142; HT/BPTB = 148) evaluated the proportion of patients with a positive pivot-shift test. A non-significant trend toward fewer positive results was observed with QT (RR = 0.57; 95% CI 0.30 to 1.08; $p = 0.08$; $I^2 = 0\%$; $p_{het} = 0.71$) (Figure 9a). Although the pooled estimate numerically favoured QT, the CI crossed unity, indicating statistical non-significance. The absence of heterogeneity

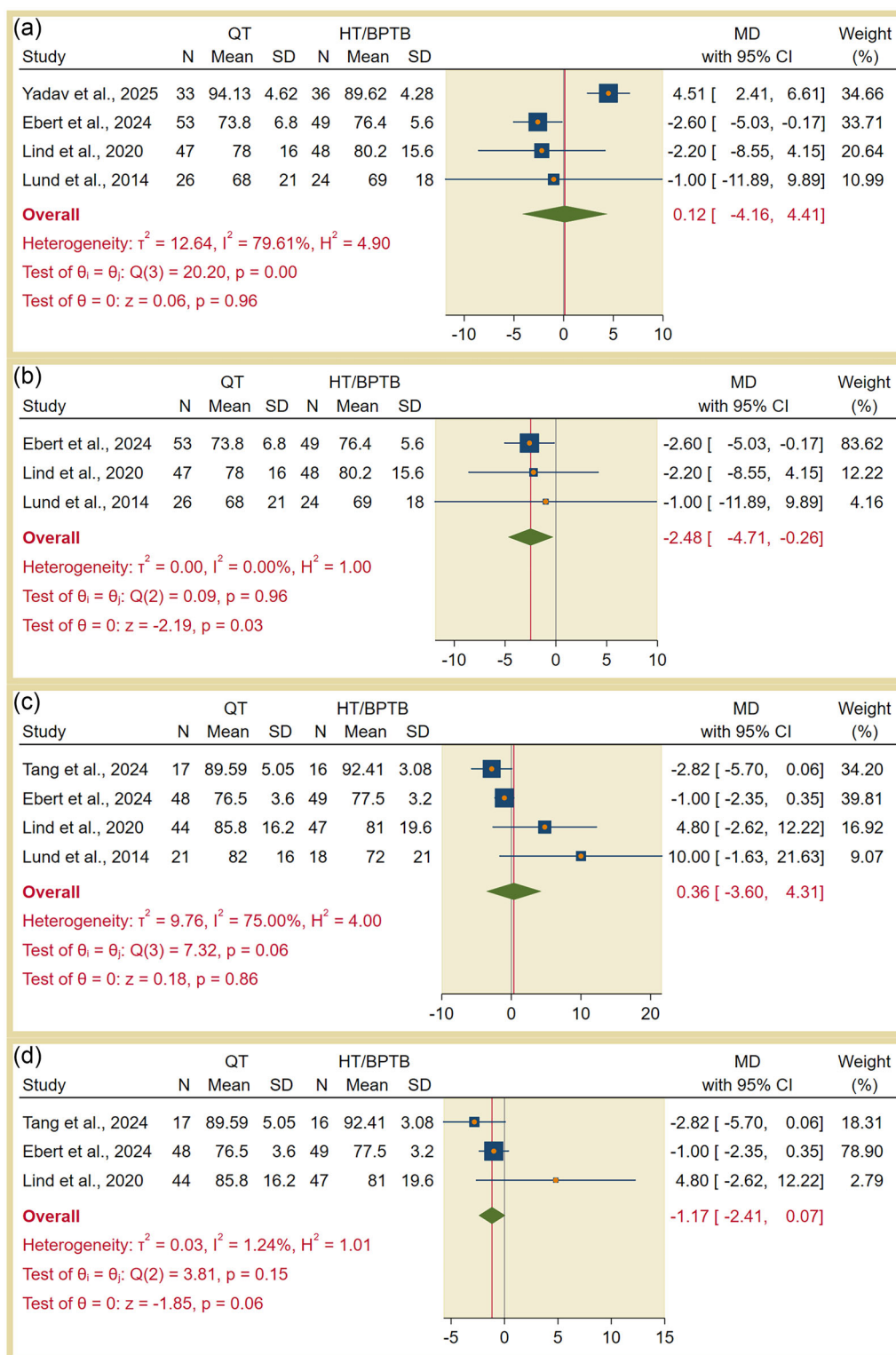


FIGURE 6 KOOS scores. Forest plots summarizing pooled analyses of KOOS across graft types. Sensitivity analyses are presented where heterogeneity was substantial. BPTB, bone–patellar tendon–bone; CI, confidence interval; HT, hamstring tendon; KOOS, Knee Injury and Osteoarthritis Outcome Score; MD, mean difference; QT, quadriceps tendon; SD, standard deviation. (a) KOOS scores at 12 months. (b) Sensitivity analysis of 12-month KOOS scores after exclusion of Yadav et al. [48]. (c) KOOS scores at 24 months. (d) Sensitivity analysis of 24-month KOOS scores after exclusion of Lund et al. [19].

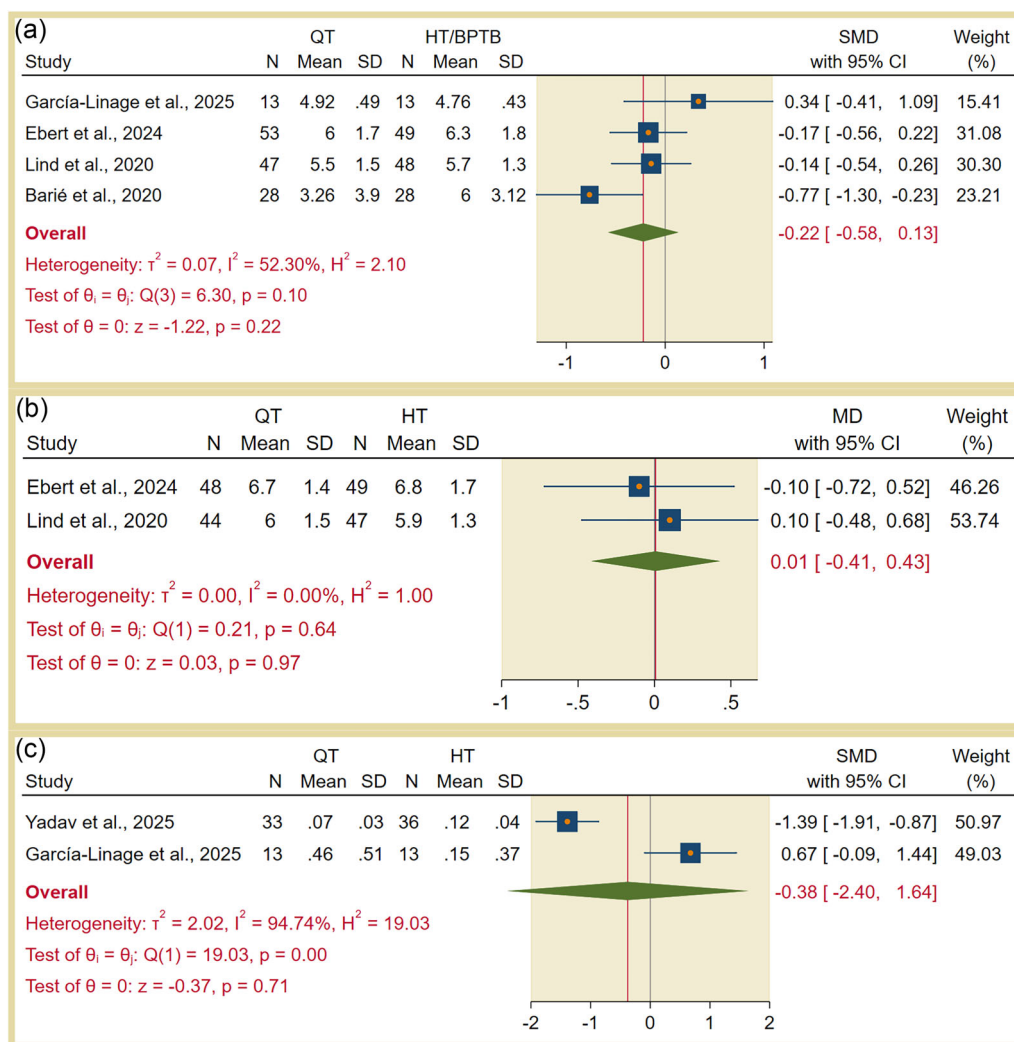


FIGURE 7 Tegner activity score and VAS pain. Forest plots evaluating postoperative activity level (Tegner scale) and pain intensity (VAS). For Tegner, higher scores represent greater activity level. For VAS, lower scores indicate less pain. BPTB, bone–patellar tendon–bone; CI, confidence interval; HT, hamstring tendon; MD, mean difference; QT, quadriceps tendon; SD, standard deviation; SMD, standardized mean difference; VAS, Visual Analogue Scale. (a) Tegner activity scores at 12 months. (b) Tegner activitiescores at 24 months. (c) VAS pain scores at 12 months.

($I^2 = 0\%$) suggests consistency across studies. Sensitivity analysis confirmed stability of the effect estimate.

Post-operative stiffness

Two studies [7, 48] (QT = 90; HT = 91) reported no significant difference in postoperative stiffness (RR = 1.02; 95% CI 0.11–9.66; $p = 0.98$; $I^2 = 0\%$; $p_{het} = 0.32$) (Figure 9b).

Results of the narrative synthesis

Calvert et al. [4] compared outcomes of ACL reconstructions using HT and QT autografts. This study found no statistically significant differences in KT-90 scores were detected between groups at either 6 or

12 months. However, by 24 months, kneeling tolerance was significantly greater in the QT group (mean HT 93 ± 9 vs. QT 98 ± 5 ; $p = 0.003$). For KT-110 measurements, the QT group outperformed the HT group at 6 (mean HT 80 ± 25 vs. QT 89 ± 12 ; $p = 0.027$), 12 (mean HT 90 ± 13 vs. QT 95 ± 10 ; $p = 0.040$) and 24 months (mean HT 92 ± 10 vs. QT 97 ± 5 ; $p = 0.003$). IKDC scores showed weak associations with KT-90 and KT-110 at 6 months, increasing to moderate correlations by 12 and 24 months. Lysholm scores were moderately correlated with both measures at most time points, whereas ACL-RSI correlations strengthened from negligible at 6 months to strong by 24 months. Global Rating of Change (GRC) scores remained weakly correlated throughout.

Postoperative complications were summarized across several studies [3, 7, 17, 19, 41, 44]. Arthrofibrosis

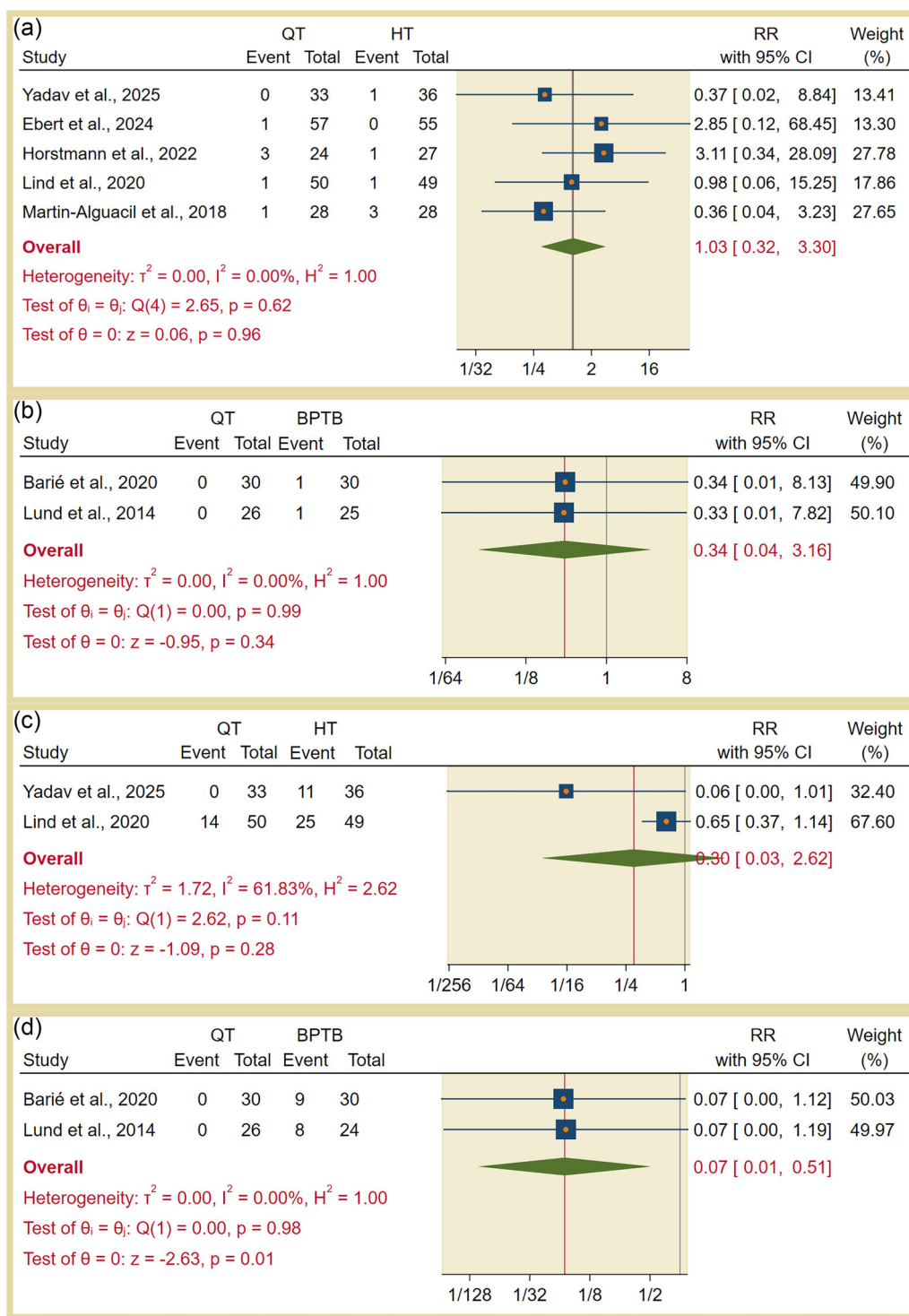


FIGURE 8 Graft failure and donor-site morbidity. Forest plots comparing: (a) QT versus HT graft failure; (b) QT versus BPTB graft failure; (c) QT versus HT donor-site morbidity; (d) QT versus BPTB donor-site morbidity. Effect measure: RR with 95% CI. RR < 1 favours QT. BPTB, bone–patellar tendon–bone; CI, confidence interval; HT, hamstring tendon; QT, quadriceps tendon; RR, risk ratio; VAS, Visual Analogue Scale.

occurred in seven QT, three BPTB and seven HT reconstructions [3, 7, 17]. Local implant irritation caused by tibial screw protrusion was reported once for each autograft type [17, 19]. Infections were documented twice with QT grafts [14, 19], while palsy of the infrapatellar branch of the saphenous nerve was reported in four HT

cases [41]. Reoperations included scar-tissue debridement in QT and BPTB groups [7], meniscal repair (two QT and four HT) [3, 7, 19], osteochondral autologous transplantation (one QT) [7], partial meniscectomy (one BPTB) [19], plica resection (one QT) [7], tibial screw removal (all graft types) [17, 19], staple removal (one HT) [7] and

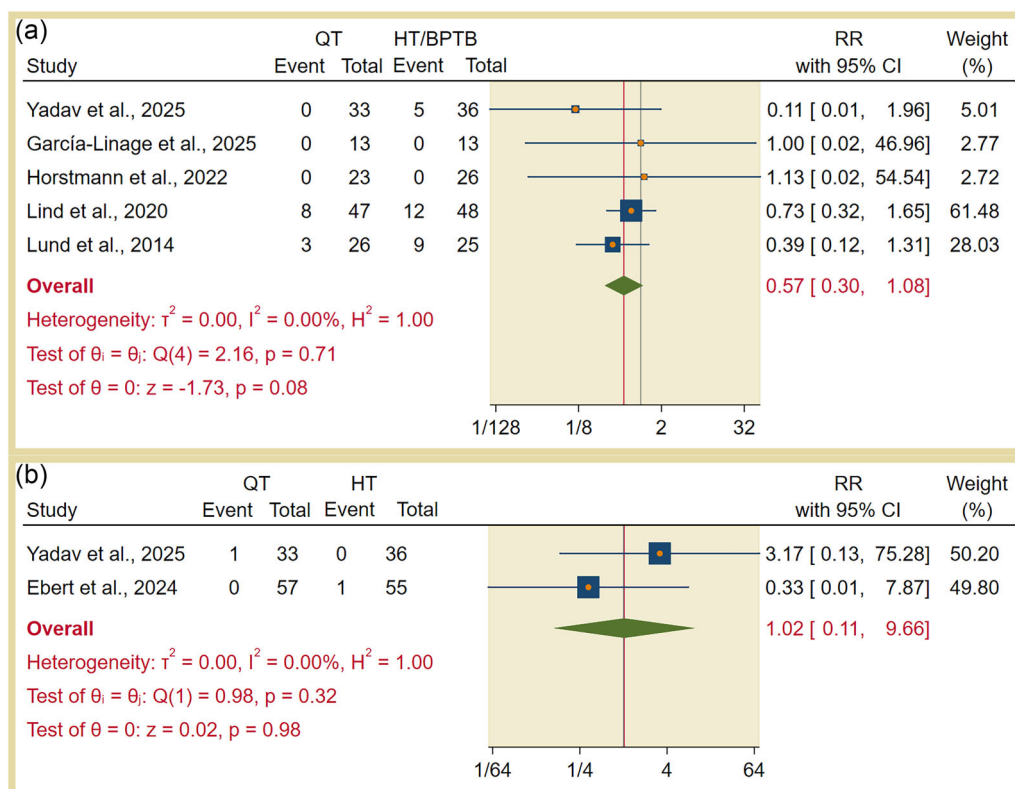


FIGURE 9 Pivot-shift test and postoperative stiffness. Forest plots assessing: (a) Proportion of positive pivot-shift tests. (b) Postoperative stiffness requiring intervention. Effect measure: RR with 95% CI. BPTB, bone–patellar tendon–bone; CI, confidence interval; HT, hamstring tendon; QT, quadriceps tendon; RR, risk ratio.

notch debridement (one QT) [7]. Owing to substantial variability in reporting, pooled meta-analysis of these complications was not feasible.

Ebert et al. [7] found that recipients of QT autografts demonstrated significantly higher hamstring limb symmetry indices (LSIs) at 6, 12 and 24 months ($p < 0.05$). Conversely, the HT group showed superior quadriceps LSIs at 6 and 12 months ($p < 0.05$). In line with this, Martin-Alguacil et al. [23] observed that QT autografts yielded a significantly higher isokinetic hamstring-to-quadriceps (HQ) ratio than HT grafts at 12 months ($p < 0.01$). Martin-Alguacil et al. [22] reported no between-group differences in pressure-pain thresholds across measured regions—including the epicondyle, vastus lateralis, vastus medialis, patellar tendon and QT (all $p > 0.05$). Ultrasonographic evaluation revealed similar QT and patellar cartilage thickness between groups. These data suggest that both QT and HT grafts provide comparable outcomes regarding pain sensitivity and soft-tissue morphology within the first postoperative year.

Sinding et al. [35] also demonstrated that QT autografts produced higher HQ ratios than HT grafts. At 1 year, patients with QT grafts showed more pronounced deficits in knee extensor strength, whereas HT graft recipients had moderate impairments in both extensors and flexors. Similarly, Tang et al. [41] found

significantly higher HQ ratios in QT reconstructions at 24 months ($p < 0.05$).

Certainty of the evidence

Overall, the certainty of evidence was mainly moderate, with downgrades primarily driven by imprecision; the risk of bias and publication bias were generally not serious/undetected. Objective stability (KT-1000) at 12 and 24 months and global function at 24 months (IKDC, Lysholm) were moderate certainty due to imprecision, while 12-month IKDC and Lysholm were low certainty because of indirectness (mixed comparators) plus imprecision. KOOS was moderate at 12 months (downgraded for indirectness) but low at 24 months (indirectness and imprecision). Activity level (Tegner) was low at 12 months (indirectness and imprecision) and moderate at 24 months (imprecision). Pain (VAS) was of low certainty owing to very serious inconsistencies with no downgrades. Donor-site morbidity was high certainty favouring QT versus BPTB and moderate certainty versus HT due to imprecision. Pivot-shift was of moderate certainty with a downgrade for indirectness, and postoperative stiffness was moderate with a downgrade for imprecision. Publication bias was undetected across outcomes (Table 3).

TABLE 3 GRADE certainty of evidence assessment.

Outcome (timepoint; comparison)	Studies (n)	Effect	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Certainty
KT-1000 side-to-side (12 months; QT vs. HT)	3 (261)	SMD -0.20 (-0.55, 0.14)	Not serious	Not serious ($I^2 = 49\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗ Moderate
KT-1000 side-to-side (24 months; QT vs. HT)	2 (144)	MD -0.15 mm (-0.51, 0.21)	Not serious	Not serious ($I^2 = 0\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗ Moderate
KT-1000 side-to-side (12 months; QT vs. BPTB)	2 (106)	MD 0.16 mm (-0.54, 0.85)	Not serious	Not serious ($I^2 = 52\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗ Moderate
IKDC (12 months; QT vs. HT/BPTB)	3 (248)	MD -1.64 (-5.15, 1.87)	Not serious	Not serious ($I^2 = 0\%$)	Serious^b	Serious^a	Undetected	⊗⊗⊗ Low
IKDC (24 months; QT vs. HT)	3 (222)	MD -1.57 (-5.01, 1.87)	Not serious	Not serious ($I^2 = 38\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗ Moderate
Lysholm (12 months; QT vs. HT/BPTB)	4 (212)	MD -0.62 (-2.73, 1.48)	Not serious	Not serious ($I^2 = 0\%$)	Serious^b	Serious^a	Undetected	⊗⊗⊗ Low
Lysholm (24 months; QT vs. HT)	3 (177)	MD -0.14 (-2.46, 2.19)	Not serious	Not serious ($I^2 = 0\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗ Moderate
KOOS (12 months; QT vs. HT/BPTB)	3 (247)	MD -2.48 (-4.71, -0.26)	Not serious	Not serious ($I^2 = 0\%$)	Serious^b	Not serious	Undetected	⊗⊗⊗ Moderate
KOOS (24 months; QT vs. HT/BPTB)	3 (221)	MD -1.17 (-2.41, 0.07)	Not serious	Not serious ($I^2 = 1.24\%$)	Serious^b	Serious^a	Undetected	⊗⊗⊗ Low
Tegner activity (12 months; QT vs. HT/BPTB)	4 (279)	SMD -0.22 (-0.58, 0.13)	Not serious	Not serious ($I^2 = 52\%$)	Serious^b	Serious^a	Undetected	⊗⊗⊗ Low
Tegner activity (24 months; QT vs. HT)	2 (188)	MD 0.01 (-0.41, 0.43)	Not serious	Not serious ($I^2 = 0\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗ Moderate
VAS pain (12 months; QT vs. HT)	2 (95)	SMD -0.38 (-2.40, 1.64)	Not serious	Very serious^c ($I^2 \approx 95\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗ Low
Graft failure (QT vs. HT)	5 (387)	RR 1.03 (0.32–3.30)	Not serious	Not serious ($I^2 = 0\%$)	Not serious	Not serious	Undetected	⊗⊗⊗⊗ High
Graft failure (QT vs. BPTB)	2 (111)	RR 0.34 (0.04–3.16)	Not serious	Not serious ($I^2 = 0\%$)	Not serious	Not serious	Undetected	⊗⊗⊗⊗ High
Donor-site morbidity (QT vs. HT)	2 (168)	RR 0.30 (0.03–2.62)	Not serious	Not serious ($I^2 = 62\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗ Moderate
Donor-site morbidity (QT vs. BPTB)	2 (110)	RR 0.07 (0.01–0.51)	Not serious	Not serious ($I^2 = 0\%$)	Not serious	Not serious (precision adequate)	Undetected	⊗⊗⊗⊗ High

(Continues)

TABLE 3 (Continued)

Outcome (timepoint; comparison)	Studies (n)	Effect	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Certainty
Pivot-shift positive (QT vs. HT/BPTB)	5 (290)	RR 0.57 (0.30–1.08)	Not serious	Not serious ($I^2 = 0\%$)	Serious^b	Not serious (precision adequate)	Undetected	⊗⊗⊗⊗ Moderate
Post-operative stiffness (QT vs. HT)	2 (181)	RR 1.02 (0.11–9.66)	Not serious	Not serious ($I^2 = 0\%$)	Not serious	Serious^a	Undetected	⊗⊗⊗⊗ Moderate

Note: Summary of the certainty of evidence for each pooled outcome according to GRADE methodology, including reasons for downgrading (risk of bias, inconsistency, indirectness, imprecision, publication bias). High certainty: we are very confident that the true effect lies close to that of the estimate of the effect. Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Abbreviations: BPTB, bone–patellar tendon–bone; CI, confidence interval; GRADE, Grading of Recommendations, Assessment, Development and Evaluation; HT, hamstring tendon; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; KT-1000, Knee laxity testing device; MD, mean difference; QT, quadriceps tendon; RR, risk ratio; SD, standard deviation; SMD, standardized mean difference; VAS, Visual Analogue Scale.

^aWide CI showing no effect.

^bDifferences in applicability of interventions.

^cModerate or substantial heterogeneity across studies.

DISCUSSION

The most important finding of this study was that QT autografts provide anterior knee stability, patient-reported outcomes and activity levels comparable to HT and BPTB autografts at 12–24 months, with lower donor-site morbidity compared with BPTB and no increase in graft failure compared with HT. Overall, these findings indicate that QT provides functional and structural outcomes equivalent to established graft options in the short- to mid-term. Regarding rotational stability, a non-significant trend toward fewer positive pivot-shift findings was observed with QT. Although the pooled estimate numerically favoured QT, the CI crossed unity, precluding definitive conclusions. Nevertheless, the direction and magnitude of effect may suggest a possible reduction in residual rotational laxity. Given the limited number of events and modest cumulative sample size, a Type II error cannot be excluded. This finding should be viewed as hypothesis-generating and warrants confirmation in adequately powered randomized trials with standardized assessment protocols.

Across the pooled analyses, the between-study heterogeneity identified at the specified endpoints is explained by discrete, table-reported methodological and comparator differences that render the implicated trials non-exchangeable with the remainder of the evidence set. Horstmann et al. [14] used an implant-free press-fit technique and a quadriceps tendon with patellar bone block (QTB), a reconstruction profile that diverged from the predominantly soft-tissue QT with conventional fixation in the comparison studies; its removal therefore reduced inconsistency in IKDC at 24 months by restoring procedural homogeneity. Yadav et al. [48] evaluated a soft-tissue QT via an all-inside approach with suspensory-button fixation and compared it to HT only. In contrast, the meta-analytic comparison pooled HT/BPTB, excluding this study, which aligned both fixation strategies and comparator composition, thereby resolving heterogeneity for Lysholm and KOOS at 12 months. Lund et al. [19] directly contrasted QTB with BPTB (bone-to-bone constructs using interference screws). They recruited patients in whom surgeons had initially selected BPTB—features that differed from trials contributing HT comparators. This approach standardized the comparator mix and graft construct, thereby eliminating heterogeneity for KOOS at 24 months. In sum, exclusion of these outliers—each distinguished by graft construct (soft-tissue QT vs. QTB), fixation method (press-fit or suspensory vs. interference screws), surgical approach (all-inside) and/or comparator composition (HT-only or BPTB-only vs. pooled HT/BPTB)—yielded a more methodologically coherent evidence set and resolved the observed inconsistency. These construct- and fixation-specific differences, including soft-tissue QT versus QTB, suspensory versus interference fixation and surgical approach variations, rendered certain studies non-exchangeable with the remainder of the evidence set. Accordingly, the pooled

analyses do not permit conclusions regarding the superiority of any specific QT technique; inferences are therefore limited to QT as a graft category.

Our meta-analysis, which shows comparable stability, function, activity and failure rates across QT, HT and BPTB at 12–24 months, aligns with the most recent pairwise and network meta-analyses. White et al. [45] found no significant differences in graft failure, laxity or patient-reported outcome measures (PROMs) at 12 and 24 months. They reported lower donor-site morbidity for QT versus both HT and BPTB, with a very low-certainty, time-limited KOOS signal favouring HT/BPTB at 12 months that dissipated thereafter—patterns concordant with our trial-only estimates when inconsistency was resolved. Dai et al. [5] and Mouarbes et al. [25] each reported equivalent failure and stability for QT versus HT and QT versus BPTB, with less donor-site pain for QT versus BPTB and small, inconsistent differences in Lysholm that did not persist across time points. Tan et al. [40] and Raj et al. [30] similarly showed no significant differences in IKDC, Lysholm, Tegner, laxity or failure between QT and HT, with a possible reduction in donor-site morbidity favouring QT that did not reach firm certainty. The network analysis by Kunze et al. [15] ranked donor-site morbidity best with QT, intermediate with HT and worst with BPTB, mirroring our high-certainty reduction in donor-site morbidity for QT versus BPTB.

QT's favourable donor-site profile versus BPTB in our RCTs likely reflects avoidance of patellar tendon violation. In contrast, equivalence versus HT reflects lower neurologic risk at the medial tibial harvest site for QT.

Although the randomized evidence included in this review was largely limited to 12–24 months of follow-up, longer-term registry studies and meta-analyses provide additional context regarding graft durability, revision risk and osteoarthritis outcomes beyond the time horizon of the included RCTs. Xie et al. [47] reported no differences between BPTB and HT in stability or function at ≥ 5 years, but higher anterior knee pain and radiographic osteoarthritic changes with BPTB. Liukkonen et al. [18], in a large pooled registry ($n = 52,878$), reported generally low revision rates, with slightly higher rates after HT compared with BPTB and increasing revision risk over time in hamstring grafts. Similarly, Rahardja et al. [29] identified younger age, hamstring grafts, smaller graft diameter and certain fixation techniques as key predictors of revision, whereas BPTB was associated with lower revision risk. Kurkowski et al. [16] reported comparable ≥ 5 -year failure rates across grafts (BPTB 6.4%, QT 9.1%, HT 12.7%), although QT remains underrepresented in long-term datasets. Overall, these findings support our primary conclusion that QT performs equivalently to HT and BPTB within randomized 24-month evidence, while observed long-term differences likely reflect surgical technique, fixation strategy and patient factors rather than intrinsic graft superiority.

Donor-site outcomes are consistent across evidence layers. Our pooled RCTs show high certainty of lower donor-site morbidity for QT versus BPTB and no apparent difference versus HT once imprecision is considered. Prior meta-analyses confirm the QT versus BPTB advantage [5, 25] and suggest equivalence versus HT [1, 30, 40]. Historical and biomechanical reviews indicate that QT has less donor-site morbidity [36] and rank it highest in this regard [15]. Conversely, long-term knee pain is more frequent with BPTB than with HT, consistent with the findings of Xie et al. [47].

Return-to-sport (RTS) data from mixed designs should be interpreted with restraint, but they are directionally coherent with our short-term RCT equivalence. DeFazio et al. [6] reported an overall RTS of 73.2%, which was higher with BPTB (81.0%) than with HT (70.6%), with similar preinjury-level RTS and re-rupture rates. Kurkowski et al. (2025) [16] found RTS at any level of 68.6% (QT), 69.9% (HT) and 55.2% (BPTB), with multiple comparisons underpowered. Our meta-analysis did not demonstrate meaningful between-graft differences in Tegner at 12–24 months, consistent with these mixed findings.

Strength and sensorimotor profiles across included and prior studies explain the clinical equivalence in stability and PROs despite harvest-specific trade-offs. Within our included trials, QT recipients showed higher hamstring-to-quadriceps ratios at 12–24 months, while HT recipients showed earlier quadriceps LSI advantages; pain sensitivity and tendon/cartilage morphology were similar [7, 22, 23, 41]. Herbawi et al. [12] and Tsai et al. [42] corroborate graft-dependent strength asymmetries, suggesting that full quadriceps recovery often lags beyond 1 year and may take up to 54–59 months post-ACLR. These patterns are directionally consistent with our finding of equivalent instrumented laxity (KT-1000) and pivot-shift across grafts, while explaining patient-level heterogeneity in early function.

Finally, practice-pattern evidence helps interpret comparator mixes and technique heterogeneity observed in our trials. Tuca et al. [43] reported worldwide preference for HT (80.3%), frequent medial-portal drilling (78.5%), cortical buttons for femoral HT fixation (82.7%), bioabsorbable tibial screws (62.7%) and a mean of 8.9 ± 2.0 months before full sports resumption; extra-articular augmentation was rare in primary ACLR (10.0%). In a nationwide study of 39,964 ACLRs, PT/QT grafts were more frequently used by experienced, high-volume surgeons and in patients with MCL injuries or those involved in pivoting sports. In contrast, female sex and mid-volume surgeons were associated with lower PT/QT use [31]. These data clarify why some RCTs featured suspensory fixation, all-inside techniques or single-graft comparators and support our sensitivity approach of separating QT versus HT and QT versus BPTB whenever feasible.

The included RCTs suggest that QT and QTB represent biologically distinct constructs; however, no clinically

meaningful differences in KT-1000 stability, pivot-shift or PROMs were observed at 12–24 months. Donor-site morbidity, in contrast, appears construct-dependent and is consistently higher with BPTB-related grafts. Overall, while biological differences between bone-to-bone and soft-tissue healing are plausible, the current randomized evidence does not demonstrate clinically relevant construct-specific superiority within the studied follow-up period [14, 19, 37].

Registry-level analyses consistently identify younger age, hamstring grafts, smaller graft diameter, antero-medial femoral drilling and suspensory fixation as predictors of higher revision risk, while BPTB and intratunnel fixation appear protective [29]. Graft diameter has also been associated with stiffness complications, with increased odds of arthrofibrosis reported in smaller grafts [39]. Overall revision rates remain low but are numerically lowest with BPTB and slightly higher with HT, with time trends suggesting a gradual increase in HT revisions [18]. Contemporary practice patterns indicate that HT remains the most frequently used graft in primary ACLR, while PT/QT usage is more common among high-volume surgeons and selected patient profiles [31, 43].

Although outside the scope of the present analysis focused on primary ACLR, revision studies suggest that QT is a viable option in revision settings, with comparable or favourable outcomes relative to HT and BPTB in terms of stability and patient-reported outcomes [2, 24]. These findings support the broader external validity of QT without altering the conclusions of this primary ACL reconstruction meta-analysis.

For adults undergoing primary ACL reconstruction, QT provides stability and patient-reported outcomes comparable to HT and BPTB at 12–24 months, with a consistent reduction in donor-site morbidity compared with BPTB and no evidence of increased graft failure compared with HT. These findings suggest that QT represents a valid alternative autograft option alongside HT and BPTB within the limits of available randomized evidence.

Shared decision-making should consider patient-specific factors such as occupational or sport-related kneeling demands, tolerance for anterior knee symptoms and surgeon experience with graft harvest and fixation techniques. Rehabilitation considerations derived from included trials may inform individualized protocols, particularly regarding early loading in suspensory-fixed QT constructs and progressive strength recovery.

Given the predominantly moderate certainty of evidence and the high certainty advantage in donor-site morbidity versus BPTB, QT may be considered an acceptable alternative graft option in appropriately selected patients. However, longer-term data beyond 5 years remain limited, and there is insufficient evidence to determine the superiority of any QT fixation or technical variant. Therefore, conclusions should be

interpreted within the constraints of short- to mid-term randomized evidence.

Strengths include a prospectively registered, PRISMA-conformant review; exclusive inclusion of randomized comparisons; duplicate screening and extraction; RoB 2 appraisal; random-effects synthesis with prespecified sensitivity analyses; separation of QT versus HT and QT versus BPTB where feasible; and GRADE certainty ratings. Heterogeneity was proactively interrogated and resolved when caused by non-exchangeable design/construct outliers, after which pooled directions remained unchanged. Most outcomes achieved moderate certainty; donor-site morbidity versus BPTB reached high certainty. Limitations reflect those of the trials, including modest sample sizes, incomplete blinding, selective reporting (domain-level ‘some concerns’), imprecision around minor effects and indirectness where HT and BPTB comparators required pooling. Follow-up seldom exceeded 24 months, constraining inferences about late failure or osteoarthritis. Future research should prioritize adequately powered, multicenter RCTs with ≥5-year follow-up to address sparse long-term QT data, using pre-registered protocols, prespecified effect sizes, blinded assessment and a harmonized core outcome set (uniform IKDC, KOOS subscales, pivot-shift grading and donor-site morbidity definitions). Trials should be construct- or fixation-specific (e.g., soft-tissue QT with suspensory fixation vs. QTB with interference screws) to separate biological from technical effects; embed standardized harm and reoperation reporting to permit pooling; and incorporate rehabilitation-embedded designs that prospectively test return-to-sport criteria anchored to strength symmetry and patient-reported readiness. Stratified analyses by age, sex, sport and graft diameter are needed to explain heterogeneity in failure/revision in line with registry risk signals. Finally, registry linkage and economic evaluations should quantify revision, contralateral injury, work re-engagement and kneeling tolerance at a scale that enables guideline-level certainty.

CONCLUSION

Across 15 randomized trials, QT, HT and BPTB autografts yield similar stability, function and activity at 12–24 months, with lower donor-site morbidity for QT compared to BPTB and no significant difference in failure rates versus HT. Inconsistency was attributed to non-exchangeable surgical constructs and comparator mixes; removing these trials restored coherence without altering the conclusions. The overall certainty is moderate, primarily limited by imprecision and short follow-up, with long-term outcomes for QT remaining insufficiently studied. For contemporary practice, QT can be offered alongside HT and BPTB as a safe and effective option.

For definitive guidance, adequately powered, construct-specific and long-term RCTs with standardized outcomes and registry linkage are the next step.

AUTHOR CONTRIBUTIONS

Mohamed Abdo Khalafallah led the conceptualization, study design, literature search, data extraction, statistical analysis, interpretation of results and drafting of the manuscript. Omar Khaled Hall contributed to data extraction, quality assessment and manuscript review. Ziad Elmenawy assisted with data extraction, methodological support and data verification. Kazhal Shekh Mohamad Ibrahim contributed to literature screening, data curation and preparation of tables and figures. Omar Mohammad Ghoneim provided methodological input, clinical interpretation and critical manuscript revisions. Abdullah Raizah and Mohamed Samir Ahmed Zaki contributed clinical expertise, supervised the interpretation of surgical outcomes and critically reviewed the manuscript. Turki Alqahtani validated the statistical methods, supervised data analysis and provided critical intellectual input. Mohamed A. Imam provided senior oversight, ensured methodological rigour, contributed to result interpretation, critically revised the manuscript, approved the final version and served as the guarantor of the work.

ACKNOWLEDGEMENTS

The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project under grant number RGP2/112/46.

FUNDING

The authors have no funding to report.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the Supporting Information of this article.

ETHICS STATEMENT

The authors have nothing to report.

ORCID

Mohamed Abdo Khalafallah  <https://orcid.org/0009-0000-1454-2589>

Ziad Elmenawy  <https://orcid.org/0000-0002-3138-9675>

REFERENCES

1. Ajrawat P, Dwyer T, Whelan D, Theodoropoulos J, Murnaghan L, Bhargava M, et al. A comparison of quadriceps tendon autograft with bone–patellar tendon–bone autograft and hamstring tendon autograft for primary anterior cruciate
- ligament reconstruction: a systematic review and quantitative synthesis. *Clin J Sport Med*. 2021;31(4):392–9.
2. Ashy C, Bailey E, Hutchinson J, Brennan E, Bailey R, Michael Pullen W, et al. Quadriceps tendon autograft has similar clinical outcomes when compared to hamstring tendon and bone–patellar tendon–bone autografts for revision ACL reconstruction: a systematic review and meta-analysis. *Knee Surg Sports Traumatol Arthrosc*. 2023;31(12):5463–76.
3. Barié A, Sprinckstubb T, Huber J, Jaber A. Quadriceps tendon vs. patellar tendon autograft for ACL reconstruction using a hardware-free press-fit fixation technique: comparable stability, function and return-to-sport level but less donor site morbidity in athletes after 10 years. *Arch Orthop Trauma Surg*. 2020;140(10):1465–74.
4. Calvert ND, Ebert JR, Radic R. Kneeling tolerance when using quadriceps tendon autograft for anterior cruciate ligament reconstruction is superior to hamstring tendon autograft. *Knee Surg Sports Traumatol Arthrosc*. 2025;33(7):2390–6.
5. Dai W, Leng X, Wang J, Cheng J, Hu X, Ao Y. Quadriceps tendon autograft versus bone–patellar tendon–bone and hamstring tendon autografts for anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Am J Sports Med*. 2021;50(12):3425–39.
6. DeFazio MW, Curry EJ, Gustin MJ, Sing DC, Abdul-Rassoul H, Ma R, et al. Return to sport after ACL reconstruction with a BTB versus hamstring tendon autograft: a systematic review and meta-analysis. *Orthop J Sports Med*. 2020;8(12):2325967120964919.
7. Ebert JR, Calvert ND, Radic R. A prospective randomized controlled trial investigating quadriceps versus hamstring tendon autograft in anterior cruciate ligament reconstruction. *Am J Sports Med*. 2024;52(3):660–9.
8. Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ*. 1997;315(7109):629–34.
9. García-Linage R, Lassard-Rosenthal J, Noval-García R, Muñoz-Madrado A, Fraind-Maya G, Palmero-Picazo J, et al. Injerto autólogo de isquiotibiales versus cuádriceps en la plastia de ligamento cruzado anterior. Estudio comparativo con enfoque al Return-To-Sport. *Acta Ortopédica Mexicana*. 2025;39(1):19–26.
10. Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction—GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol*. 2011;64(4):383–94.
11. Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ*. 2008;336(7650):924–6.
12. Herbawi F, Lozano-Lozano M, Lopez-Garzon M, Postigo-Martin P, Ortiz-Comino L, Martin-Alguacil J, et al. A systematic review and meta-analysis of strength recovery measured by isokinetic dynamometer technology after anterior cruciate ligament reconstruction using quadriceps tendon autografts vs. hamstring tendon autografts or patellar tendon autografts. *Int J Environ Res Public Health*. 2022;19(11):6764.
13. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al., editors. *Cochrane handbook for systematic reviews of interventions*. Chichester (UK): Wiley; 2019.
14. Horstmann H, Petri M, Tegtbu U, Felmet G, Krettek C, Jagodzinski M. Quadriceps and hamstring tendon autografts in ACL reconstruction yield comparably good results in a prospective, randomized controlled trial. *Arch Orthop Trauma Surg*. 2022;142(2):281–9.
15. Kunze KN, Moran J, Polce EM, Pareek A, Strickland SM, Williams RJ. Lower donor site morbidity with hamstring and quadriceps tendon autograft compared with bone–patellar tendon–bone autograft after anterior cruciate ligament

- reconstruction: a systematic review and network meta-analysis of randomized controlled trials. *Knee Surg Sports Traumatol Arthrosc.* 2023;31(8):3339–52.
16. Kurkowski SC, Thimmesch MJ, Murphy M, Kuechly HA, Emmert AS, Grawe B. Uncovering the state of current data on quadriceps tendon autograft use versus bone–patellar tendon–bone and hamstring tendon autografts in anterior cruciate ligament reconstruction at ≥ 5 years after surgery: a systematic review and meta-analysis. *Am J Sports Med.* 2025;53(7):1739–49.
 17. Lind M, Nielsen TG, Soerensen OG, Mygind-Klavsen B, Faunø P. Quadriceps tendon grafts does not cause patients to have inferior subjective outcome after anterior cruciate ligament (ACL) reconstruction than do hamstring grafts: a 2-year prospective randomised controlled trial. *Br J Sports Med.* 2020;54(3):183–7.
 18. Liukkonen RJ, Ponkilainen VT, Reito A. Revision rates after primary ACL reconstruction performed between 1969 and 2018: a systematic review and metaregression analysis. *Orthop J Sports Med.* 2022;10(8):23259671221110191, <https://doi.org/10.1177/23259671221110191>
 19. Lund B, Nielsen T, Faunø P, Christiansen SE, Lind M. Is quadriceps tendon a better graft choice than patellar tendon? A prospective randomized study. *Arthroscopy.* 2014;30(5):593–8.
 20. Makhmalbaf H. ACL deficient knee, an approach to patient selection and reconstruction. *Arch Bone Joint Surg.* 2013;1(2):45–7.
 21. Maletis GB, Chen J, Inacio MCS, Funahashi TT. Age-related risk factors for revision anterior cruciate ligament reconstruction: a cohort study of 21,304 patients from the Kaiser Permanente Anterior Cruciate Ligament Registry. *Am J Sports Med.* 2016;44(2):331–6.
 22. Martin-Alguacil JL, Arroyo-Morales M, Luis martin-Gómez J, Lozano-Lozano M, Galiano-Castillo N, Cantarero-Villanueva I. Comparison of knee sonography and pressure pain threshold after anterior cruciate ligament reconstruction with quadriceps tendon versus hamstring tendon autografts in soccer players. *Acta Orthop Traumatol Turc.* 2019;53(4):260–5.
 23. Martin-Alguacil JL, Arroyo-Morales M, Martín-Gomez JL, Monje-Cabrera IM, Abellán-Guillén JF, Esparza-Ros F, et al. Strength recovery after anterior cruciate ligament reconstruction with quadriceps tendon versus hamstring tendon autografts in soccer players: a randomized controlled trial. *Knee.* 2018;25(4):704–14.
 24. Meena A, Di Paolo S, Grassi A, Raj A, Farinelli L, Hoser C, et al. No difference in patient reported outcomes, laxity, and failure rate after revision ACL reconstruction with quadriceps tendon compared to hamstring tendon graft: a systematic review and meta-analysis. *Knee Surg Sports Traumatol Arthrosc.* 2023;31(8):3316–29.
 25. Mouarbes D, Menetrey J, Marot V, Courtot L, Berard E, Cavaignac E. Anterior cruciate ligament reconstruction: a systematic review and meta-analysis of outcomes for quadriceps tendon autograft versus bone–patellar tendon–bone and hamstring-tendon autografts. *Am J Sports Med.* 2019;47(14):3531–40.
 26. Mustafa RA, Santesso N, Brozek J, Akl EA, Walter SD, Norman G, et al. The GRADE approach is reproducible in assessing the quality of evidence of quantitative evidence syntheses. *J Clin Epidemiol.* 2013;66(7):736–42.e5.
 27. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ.* 2021;372:n71.
 28. Persson A, Fjeldsgaard K, Gjertsen J-E, Kjellsen AB, Engebretsen L, Hole RM, et al. Increased risk of revision with hamstring tendon grafts compared with patellar tendon grafts after anterior cruciate ligament reconstruction: a study of 12,643 patients from the Norwegian Cruciate Ligament Registry, 2004–2012. *Am J Sports Med.* 2014;42(2):285–91.
 29. Rahardja R, Zhu, M, Love, H, Clatworthy, M.G, Monk, A.P, Young, S.W. Factors associated with revision following anterior cruciate ligament reconstruction: a systematic review of registry data. *Knee.* 2020;27(2):287–99.
 30. Raj S, Ridha A, Searle H.K.C, Khatri C, Ahmed I, Metcalfe A, et al. Quadriceps tendon versus hamstring tendon graft for primary anterior cruciate ligament reconstruction: a systematic review and meta-analysis of randomised trials. *Knee.* 2024;49:226–40.
 31. Rizvanovic D, Waldén M, Forssblad M, Ståhlman A. Surgeon's experience, sports participation and a concomitant MCL injury increase the use of patellar and quadriceps tendon grafts in primary ACL reconstruction: a nationwide registry study of 39,964 surgeries. *Knee Surg Sports Traumatol Arthrosc.* 2022;31(2):475–86.
 32. Samuelsen BT, Webster KE, Johnson NR, Hewett TE, Krych AJ. Hamstring autograft versus patellar tendon autograft for ACL reconstruction: is there a difference in graft failure rate? A meta-analysis of 47,613 patients. *Clin Orthop Relat Res.* 2017;475(10):2459–68.
 33. Servant C. Editorial commentary: quadriceps tendon anterior cruciate ligament graft shows advantages and disadvantages. *Arthroscopy.* 2024;40(1):146–8.
 34. Shani RH, Umpierrez E, Nasert M, Hiza EA, Xerogeanes J. Biomechanical comparison of quadriceps and patellar tendon grafts in anterior cruciate ligament reconstruction. *Arthroscopy.* 2016;32(1):71–5.
 35. Sinding KS, Nielsen TG, Hvid LG, Lind M, Dalgas U. Effects of autograft types on muscle strength and functional capacity in patients having anterior cruciate ligament reconstruction: a randomized controlled trial. *Sports Med.* 2020;50(7):1393–403.
 36. Slone HS, Romine SE, Premkumar A, Xerogeanes JW. Quadriceps tendon autograft for anterior cruciate ligament reconstruction: a comprehensive review of current literature and systematic review of clinical results. *Arthroscopy.* 2015;31(3):541–54.
 37. Solie B, Monson J, Larson C. Graft-specific surgical and rehabilitation considerations for anterior cruciate ligament reconstruction with the quadriceps tendon autograft. *Int J Sports Phys Ther.* 2023;18(2):493–512.
 38. Sterne JAC, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ.* 2019;366:l4898.
 39. Su AW, Storey EP, Lin SC, Forst B, Lawrence JT, Ganley T.J, et al. Association of the graft size and arthrofibrosis in young patients after primary anterior cruciate ligament reconstruction. *J Am Acad Orthop Surg.* 2018;26(23):e483–9.
 40. Tan TK, Subramaniam AG, Ebert JR, Radic R. Quadriceps tendon versus hamstring tendon autografts for anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Am J Sports Med.* 2021;50(14):3974–86.
 41. Tang N, Eren M, Gurpinar T, Ozturkmen Y. A prospective randomized controlled study of hamstring and bone-free quadriceps tendons autografts in arthroscopic ACL reconstruction. *Eur J Orthop Surg Traumatol.* 2024;34(1):293–301.
 42. Tsai LC, Jeanfreau CM, Hamblin KA, Popovich JM, Lyle MA, Cottmeyer DF, et al. Time, graft, sex, geographic location, and isokinetic speed influence the degree of quadriceps weakness after anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Knee Surg Sports Traumatol Arthrosc.* 2022;30(10):3367–76.
 43. Tuca M, Valderrama I, Eriksson K, Tapasvi S. Current trends in anterior cruciate ligament surgery. A worldwide benchmark study. *J ISAKOS.* 2023;8(1):2–10.
 44. Vilchez-Cavazos F, Dávila-Martínez A, Garza-Castro S, Simental-Mendía M, Garay-Mendoza D, Tamez-Mata Y, et al. Anterior cruciate ligament injuries treated with quadriceps tendon autograft versus hamstring autograft: a randomized controlled trial. *Cir Cir.* 2020;88(1):76–81.

45. White T, Castro M, Antonio L, Hing W, Tudor F, Sattler L. Quadriceps, hamstring and patella tendon autografts for primary anterior cruciate ligament reconstruction demonstrate similar clinical outcomes, including graft failure, joint laxity and complications: a systematic review with meta-analysis of randomised controlled trials. *Knee Surg Sports Traumatol Arthrosc.* 2026;34(5):1631–46.
46. Xie X, Liu X, Chen Z, Yu Y, Peng S, Li Q. A meta-analysis of bone–patellar tendon–bone autograft versus four-strand hamstring tendon autograft for anterior cruciate ligament reconstruction. *Knee.* 2015;22(2):100–10.
47. Xie X, Xiao Z, Li Q, Zhu B, Chen J, Chen H, et al. Increased incidence of osteoarthritis of knee joint after ACL reconstruction with bone–patellar tendon–bone autografts than hamstring autografts: a meta-analysis of 1,443 patients at a minimum of 5 years. *Eur J Orthop Surg Traumatol.* 2015;25(1):149–59.
48. Yadav AK, Sharma A, Paul N, Bhakhar A, Sarkar B, Azam MQ. Quadriceps versus hamstrings autograft in anterior cruciate ligament reconstruction: a randomized control trial. *Eur J Orthop Surg Traumatol.* 2025;35(1):395.
49. Zhang X-F, Liu P, Huang J-W, He Y-H. Efficacy and safety of quadriceps tendon autograft versus bone–patellar tendon–bone and hamstring tendon autografts for anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *J Orthop Traumatol.* 2024;25(1):65.
50. Zhu H, Tseng F, Lee K, Ho SWL, Tan LTJ, Jegathesan T. Quadriceps tendon autograft: a larger intraoperative diameter for primary anterior cruciate ligament reconstruction in Asian patients. *J Arthrosc Surg Sports Med.* 2026;7:56–62.
51. Rayyan—intelligent systematic review. Rayyan; 2021.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Khalafallah MA, Hall OK, Elmenawy Z, Ibrahim KSM, Ghoneim OM, Raizah A, et al. Quadriceps tendon autograft provides comparable stability and functional outcomes with lower donor-site morbidity than bone–patellar tendon–bone: a systematic review and meta-analysis of 15 randomized controlled trials with GRADE evidence. *Knee Surg Sports Traumatol Arthrosc.* 2026;1–25.

<https://doi.org/10.1002/ksa.70499>